Shifting Comparative Advantage and the Value of the World Trading System

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Abstract

This paper measures how shifting comparative advantage affects the value of the world trading system. In a Ricardian model, the system restrains tariffs through cooperation (concern for foreign welfare) and commitments (tariff ceilings). In the 2017 baseline, governments weigh foreign welfare at 47 percent of their own, as revealed by tariffs they are free to adjust. Removing both restraints lowers world real income by 0.81 percent. Either alone protects most of this value. Larger economies generally value cooperation more, and smaller economies value commitments more. The system's value is recomputed in three counterfactual analyses. Widening comparative advantage raises aggregate value, and narrowing lowers it slightly. A shock's effect depends on which countries and sectors are affected and how trade and tariffs adjust, not on its sign alone. Across these analyses, commitments carry most of the value's response. The projected climate change scenario leaves the system's aggregate value almost unchanged.

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1 Introduction

How do changes in comparative advantage affect the welfare value of the world trading system? The question is timely, since the system has come under pressure while comparative advantage continues to shift. The current trading system restrains tariffs through cooperation and commitments. Cooperation leads governments to account for some of the costs their tariffs impose abroad. Commitments limit tariff choices through WTO bindings, substantially expanded during the Uruguay Round, and preferential tariff schedules. Productivity shifts can change trade patterns and governments' incentives to restrict trade, and therefore what each restraint is worth.

This paper adapts the revealed-preference framework for cooperation in Adão et al. (2026) by representing tariff commitments separately and evaluating how productivity changes affect the value of both restraints. The analysis uses a multi-sector Ricardian model calibrated to 2017. At each productivity state, the system's value is the welfare gain from maintaining both restraints relative to noncooperation with neither. Governments set tariffs in every regime and productivity state, each taking the others' tariffs as given. Regimes with one restraint show what that restraint protects on its own.

Low observed tariffs can reflect both concern for foreign welfare and restraint imposed by ceilings. Treating a tariff at its ceiling as an unconstrained choice can attribute the ceiling's restraint to greater cooperation. The ceilings are constructed from WTO bindings and preferential schedules. The weight on foreign welfare, assumed common to all governments and partners, is therefore estimated from tariffs classified as interior, which governments can adjust in either direction. The least-squares estimate is 0.47, compared with 0.83 when interior equality conditions are applied to the broader sample. This difference is consistent with ceilings accounting for part of the observed restraint, although sample composition also matters.

In the solved baseline, the system is worth 0.81 percent of world real income and 1.9 percent for the median economy relative to noncooperation. Cooperation alone delivers 94 percent of the system's value and commitments alone 92 percent. The restraints are therefore substitutes at the estimated weight. Commitments are generally more valuable to smaller economies through protected market access. Larger economies tend to benefit relatively more from cooperation, consistent with a common weight on the level of partners' real income, which counts a large economy's losses more.

Three counterfactual analyses then trace how productivity shifts affect the existing system's value and how each restraint responds. Theory does not sign this effect in a multi-country and multi-sector setting. It does show that a common proportional change in productivity leaves the value unchanged (Corollary 1), so only relative shifts matter. The effect of such a shift depends on where it lands relative to the trade the system protects and on how tariffs adjust. The analyses combine systematic shifts in comparative advantage, two historically scaled shocks of opposite signs in large economies, and a projected climate change scenario. Across the analyses, the system's value changes mainly through commitments, so the two restraints, nearly interchangeable in the baseline, differ in how they respond.

The first analysis widens and narrows comparative advantage while holding each economy's mean log productivity fixed, a shift scaled to the convergence documented by Levchenko and Zhang (2016). Widening raises the system's value by 2.2 percent, and the reverse shift lowers it by 0.1 percent. At fixed tariffs, wider productivity differences raise the gains from specialization, and the system's low tariffs let economies capture more of these gains than noncooperative tariffs would. Tariff adjustment partly offsets this increase, with the opposite pattern under the reverse shift.

The second analysis applies two shocks, contracting Chinese manufacturing productivity by 35 percent and expanding United States extraction productivity by 27 percent.[1] The shocks have opposite signs, yet both raise the system's value, by about 7 and 3 percent, while lowering it for most partners and for the median economy. What the system is worth to an economy depends on how much it stands to lose from a trade war. The Chinese contraction lowers China's real income but makes it depend on imported manufactures that noncooperative tariffs would tax. The extraction expansion raises United States real income but leaves it with more output to sell in export markets that noncooperative tariffs would nearly close. Each shocked economy is large, so its higher valuation of the system outweighs the net decline for its partners, whose trade with it shrinks or is displaced. A shock's sign alone therefore does not determine the response.

The third analysis applies the climate productivity projections of Nath (2025). The largest changes fall on agriculture, whereas much of the trade the ceilings protect is

[1]The Chinese contraction approximates a reversal of the manufacturing productivity growth estimated by Hsieh and Ossa (2016) for 1995–2007. The United States expansion matches growth in mining total factor productivity over 2007–2017 reported by the Bureau of Labor Statistics. Appendix D.2 provides the calculations.

advanced-economy manufacturing. At fixed tariffs, the shock lowers the gains the system protects slightly, and tariff adjustment offsets part of that decline. The system adds no more to world real income under climate change than it already adds in the baseline, although its gains are redistributed across economies.

The main findings survive sensitivity checks. The aggregate signs of the counterfactual shifts hold at the endpoints of the foreign welfare weight's confidence interval, and most survive alternative shock constructions. A tighter definition of interior tariffs lowers the estimated weight, so the division of value between cooperation and commitments depends on that definition.

The paper builds on research connecting comparative advantage and trade policy. Johnson (1953) characterizes the tariff war that noncooperation produces, Opp (2010) shows in a two-country Ricardian model that wider comparative advantage raises noncooperative tariffs, and Costinot et al. (2015) characterize optimal policy with comparative advantage. Productivity convergence can raise the welfare cost of noncooperation (Felbermayr et al., 2013), and productivity asymmetry affects agreement formation (Duan and Suwanprasert, 2026). Governments adjust agricultural trade policy to climate-driven productivity shifts (Hsiao et al., 2026). Here, the existing restraints are held fixed while productivity changes, and the analysis measures the resulting change in their value.

The two restraints each follow the trade-agreements literature. Cooperation is the internalization of terms-of-trade externalities that agreements sustain (Bagwell and Staiger, 1999; Maggi, 1999), which Nicita et al. (2018) detect within tariff water and attribute to the fear of retaliation. Commitments are tariff caps that leave applied rates free below them (Amador and Bagwell, 2013; Beshkar et al., 2015). Quantitative work infers cooperation from foreign welfare weights without modeling the caps (Adão et al., 2026). This paper keeps the caps as constraints, identifies the weight from tariffs below them, and values each restraint separately. The climate application complements research on adaptation through trade and reallocation (Costinot et al., 2016; Gouel and Laborde, 2021; Cruz and Rossi-Hansberg, 2024; Nath, 2025) by letting tariffs adjust within the existing restraints.

Section 2 presents the framework, Section 3 the data and estimation, and Section 4 the baseline values. Sections 5 and 6 report the productivity counterfactuals and the sensitivity checks. Section 7 concludes. Appendices provide derivations, data construction,

and additional results.

2 Framework

This section sets out the environment, the governments' objectives, and the regimes, and defines the value of the trading system at each productivity state. It then shows that only relative productivity changes move the value and decomposes their effect into exposure at fixed tariffs and the tariff response.

2.1 Institutions, regimes, and the value of the system

The environment is the multi-sector Ricardian model of Eaton and Kortum (2002), with sectors as in Caliendo and Parro (2015) and labor as the only input. It has N countries and S tradable sectors, indexed by $i, j \in \{1, \ldots, N\}$ and $s \in \{1, \ldots, S\}$. Each country has a fixed endowment of homogeneous labor, mobile across domestic sectors and immobile internationally. Productivities are Fréchet with country-sector scale T_{js} and sector-specific shape θ_s. Let $\mathbf{T} = (T_{js})$ denote the technology profile, and let $a_{js} = T_{js}^{1/\theta_s}$ index country-sector productivity. Appendix A states preferences, trade costs, the competitive equilibrium, and its hat-algebra representation.

Country i imposes an ad-valorem tariff $t_{ji}^s \geq 0$ on imports from j in sector s, in addition to iceberg trade costs. Write $\mathbf{t} = (\mathbf{t}_1, \ldots, \mathbf{t}_N)$, where $\mathbf{t}_i$ collects government i's tariff choices. Trade is balanced and tariff revenue is rebated. Equilibrium real income is $W_i(\mathbf{T}, \mathbf{t}) = I_i/P_i$, where I_i is the sum of labor income and tariff revenue and P_i is the consumption price index. Superscript 0 denotes the baseline allocation. Multiplication by the fixed baseline price index P_i^0 expresses real income in monetary units at baseline prices.

Following Adão et al. (2026), government i maximizes

$$G_i(\mathbf{T}, \mathbf{t}; \boldsymbol{\lambda}) = P_i^0 W_i(\mathbf{T}, \mathbf{t}) + \sum_{j \neq i} \lambda_{ij} P_j^0 W_j(\mathbf{T}, \mathbf{t}), \qquad \lambda_{ij} \geq 0, \tag{1}$$

where λ_{ij} is government i's weight on country j's welfare and $\boldsymbol{\lambda}$ collects these weights across governments. The present application represents tariff ceilings separately and holds both welfare weights and ceilings fixed when productivity changes. The distinction

Table 1: Institutional regimes

Regime	Foreign welfare weights	Feasible tariffs
System, sys	Estimated $\boldsymbol{\lambda}$	$\mathcal{D}_i^I$
Commitments only, com	Zero	$\mathcal{D}_i^I$
Cooperation only, coop	Estimated $\boldsymbol{\lambda}$	$\mathcal{D}_i$
Noncooperative, nc	Zero	$\mathcal{D}_i$

Notes: The weights are estimated in Section 3 and held fixed thereafter. The domain $\mathcal{D}_i$ excludes institutional ceilings but retains the noninstitutional limits described in Section 3.1, and $\mathcal{D}_i^I$ adds the ceilings in (2).

between the two restraints is operational. A ceiling restrains a tariff whatever the weight, and the weight measures restraint that no ceiling enforces. This is comparative statics on the existing institutions rather than a model of their renegotiation. It asks what the restraints already in place are worth once comparative advantage shifts. Bound rates have changed little for most members since the Uruguay Round, and raising them requires compensation under Article XXVIII. Preferential rates that also serve as ceilings change as agreements are signed, so I also report the results with only the WTO bindings as ceilings in Section 6 and Appendix Table E1.

Without institutional ceilings, government i chooses tariffs from a nonempty closed domain $\mathcal{D}_i$ of nonnegative tariff vectors. Commitments restrict its choices to

$$\mathcal{D}_i^I = \{\mathbf{t}_i \in \mathcal{D}_i : t_{i\ell} \leq \bar{t}_{i\ell} \text{ for each covered tariff } \ell\}. \tag{2}$$

Here ℓ indexes the elements of $\mathbf{t}_i$, and $\bar{t}_{i\ell}$ is the institutional ceiling on $t_{i\ell}$. A covered tariff is subject to a ceiling, which may be slack or binding. Uncovered tariffs face only the restrictions in $\mathcal{D}_i$. The set $\mathcal{D}_i^I$ is assumed nonempty. In the quantitative model, government i's choice variables are its bilateral tariffs by sector in every regime, so the regimes differ only in the weights and the ceilings.

A policy equilibrium is a feasible tariff profile at which each government maximizes its objective, taking other governments' tariffs as given. Each government accounts for the competitive-equilibrium responses of wages, expenditures, and tariff revenue. Table 1 defines the four regimes. Appendix B.1 gives sufficient existence conditions. Global uniqueness is not established, but Appendix B.3 shows that the baseline and climate comparisons are stable across starting points.

For each regime R, let $\mathbf{t}^R(\mathbf{T})$ denote a selected policy equilibrium and $W_i^R(\mathbf{T}) =$

$W_i(\mathbf{T}, \mathbf{t}^R(\mathbf{T}))$ its associated real income. Let $\mathbf{T}^0$ and $\mathbf{T}^1$ denote the baseline and counterfactual technology profiles. Measure the system's value by the log welfare gap and its change:

$$v_i(\mathbf{T}) = \log W_i^{\text{sys}}(\mathbf{T}) - \log W_i^{\text{nc}}(\mathbf{T}), \qquad \Delta v_i = v_i(\mathbf{T}^1) - v_i(\mathbf{T}^0). \tag{3}$$

The associated proportional welfare gain is $\exp(v_i) - 1$. Both regimes face the same technology profile in each comparison, so Δv_i measures how productivity changes the gain from maintaining the same institutions.

Two further comparisons isolate each restraint. The value of commitments alone is $v_i^{\text{com}}(\mathbf{T}) = \log W_i^{\text{com}}(\mathbf{T}) - \log W_i^{\text{nc}}(\mathbf{T})$, the gain from imposing ceilings on noncooperative governments, and the value of cooperation alone is $v_i^{\text{coop}}(\mathbf{T}) = \log W_i^{\text{coop}}(\mathbf{T}) - \log W_i^{\text{nc}}(\mathbf{T})$, the gain from the weights without ceilings. Each restraint can protect gains the other also protects, so the two values need not sum to $v_i(\mathbf{T})$, and neither need be positive.

2.2 Productivity exposure and tariff responses

Only differences in productivity change across countries or sectors move the value. Suppose productivity a_{js} rises by the same percentage in every country and sector. Relative production costs and equilibrium tariffs remain unchanged, and real income rises by the same percentage under the system and noncooperation. The percentage welfare gain from maintaining the system is therefore unchanged, although the gain in levels rises with productivity. This is why value is measured in proportional terms.[2]

These differences can change both trade exposure and governments' tariff incentives. To separate these effects, let a scalar z index a productivity path $\mathbf{T}(z)$, holding all other primitives fixed, and define $u_i(z, \mathbf{t}) = \log W_i(\mathbf{T}(z), \mathbf{t})$. The subscript z denotes a partial derivative at fixed tariffs. The operator $\nabla_{\mathbf{t}_j}$ denotes the gradient with respect to government j's tariff vector. Along differentiable policy-equilibrium branches, the chain

[2]Corollary 1 in Appendix A.4 records this standard homogeneity property for the present environment, with its tariff ceilings and foreign welfare weights, and pairs each equilibrium with its scaled counterpart when several exist.

rule gives

$$\frac{dv_i}{dz} = \underbrace{u_{i,z}(z, \mathbf{t}^{\text{sys}}) - u_{i,z}(z, \mathbf{t}^{\text{nc}})}_{\text{difference in fixed-policy exposure}} + \underbrace{\sum_{j \neq i} \left[\nabla_{\mathbf{t}_j} u_i(z, \mathbf{t}^{\text{sys}}) \cdot \frac{d\mathbf{t}_j^{\text{sys}}}{dz} - \nabla_{\mathbf{t}_j} u_i(z, \mathbf{t}^{\text{nc}}) \cdot \frac{d\mathbf{t}_j^{\text{nc}}}{dz} \right]}_{\text{partners' policy responses}} + \underbrace{\nabla_{\mathbf{t}_i} u_i(z, \mathbf{t}^{\text{sys}}) \cdot \frac{d\mathbf{t}_i^{\text{sys}}}{dz} - \nabla_{\mathbf{t}_i} u_i(z, \mathbf{t}^{\text{nc}}) \cdot \frac{d\mathbf{t}_i^{\text{nc}}}{dz}}_{\text{own policy responses}}. \tag{4}$$

The first term is the exposure term, which compares the productivity shock under the two regimes' tariff schedules, holding each schedule fixed while trade and wages adjust. The second and third terms together form the policy term, the additional effect of tariff adjustment by partners and by the government itself. Exposure and policy effects can reinforce or offset each other. The identity is a way to organize counterfactual results. It does not supply a sign prediction for a heterogeneous shock.

The two restraints enter these responses differently. A tariff that remains at a fixed ceiling cannot adjust through that coordinate. An interior tariff can move as the government balances domestic and weighted foreign welfare. At an interior noncooperative optimum, the own-policy welfare gradient is zero. Under cooperation, the domestic gradient generally remains nonzero because the government also values foreign welfare. Appendix B.2 gives these conditions and defines the finite decomposition.

3 Data and Estimation

The quantitative implementation combines a 2017 allocation of trade and applied tariffs, institutional tariff ceilings, and a foreign welfare weight estimated from tariffs governments are free to adjust. The purpose is to discipline the policy regimes before changing productivity. Following Adão et al. (2026), the weight is estimated by least squares from the relation between domestic and foreign welfare margins, with a free-trade instrument as a check.

3.1 Calibration, tariff constraints, and sample construction

This subsection constructs the baseline economy, tariff constraints, and estimation sample. The 2017 calibration covers 34 economies and a rest-of-world aggregate, treating the European Union as one economy. Production and trade are grouped into agriculture, extraction, and manufacturing to match the productivity inputs. Bilateral trade at the six-digit product level comes from BACI, and applied tariffs come from WITS. Production data are from FAOSTAT for agriculture and from UNIDO for extraction and manufacturing, supplemented by OECD observations for some missing country-sector pairs. Sectoral trade elasticities are calibrated using adapted estimates from Caliendo and Parro (2015). Appendix C.1 provides more details on the sources and parameters.

Counterfactual equilibria are computed using exact hat algebra following Dekle et al. (2008). This expresses equilibrium conditions in relative changes using baseline trade and expenditure shares, tariffs, incomes, and trade elasticities. Technology levels and iceberg costs are absorbed in the baseline shares. Recorded trade deficits are first removed at unchanged technologies and observed tariffs. The resulting balanced-trade allocation supplies the baseline incomes, shares, and prices P_i^0 (Appendix C.2).

The policy game of Section 2.1 has 32 optimizing governments and 2,989 bilateral-sector tariff choices. The bilateral specification accommodates preferential trade arrangements.[3] Saudi Arabia and the United Arab Emirates, which lack consolidated WTO schedules (Appendix C.3), and the rest of world retain observed tariffs in every regime while remaining in the competitive equilibrium.

Institutional ceilings combine WTO bindings with preferential tariff schedules. Product-level bound rates are mapped to applied tariff codes, and bound and applied rates are averaged using the same fixed baseline import weights. Preferential ceilings equal observed preferential rates, so preferential cells lie outside the estimation sample, and constructed ceilings are no lower than observed applied rates. The numerical upper bound on each tariff is the larger of its observed rate and 200 percent. No numerical upper bound binds in the reported counterfactuals. I examine the removal of preferential ceilings in Section 6 and alternative numerical caps in Appendix Table E2. Appendix C.3 details schedule

[3] In the baseline each bilateral tariff is a separate choice, so a government may treat nonpreferential partners differently. A most-favored-nation (MFN) sensitivity instead moves all of an importer's nonpreferential tariffs in a sector by one common amount, preserving their observed differences. Appendix B.1 shows how the corresponding marginal welfare effects are aggregated, and the baseline value and the climate response change little under this specification (Appendix Table E2).

coverage, missing bindings, and concordance adjustments.

The estimation sample contains strictly positive applied tariffs at least five percentage points below their constructed ceilings. These tariffs can move in either direction within the sectoral model. Such observations help identify cooperation separately from restraint imposed by ceilings. The sample carries 14 percent of active imports, since the majority of trade is on tariffs at zero or at their ceilings. The five-point margin allows for concordance and aggregation differences, and I examine two- and ten-point margins in Appendix Table E2.

Each cell's applied tariff is an import-weighted average of product tariffs that reproduces the cell's baseline tariff revenue (Appendix C.5), so an interior cell can still contain products at zero and products at their bindings. A modeled increase in the cell's tariff must then come from products with room below their bindings, and a decrease from products with positive rates. I assess this aggregation with samples of higher interior coverage in Section 3.3 and a product-level approximation in Appendix C.5, without claiming equivalence to a product-level policy model.

3.2 Estimating foreign welfare weights

The first-order condition of (1) for an interior tariff choice $t_{i\ell}$, evaluated at observed tariffs where $P_i^0 W_i^0 = I_i^0$, is

$$I_i^0 \frac{d \log W_i}{dt_{i\ell}} + \sum_{j \neq i} \lambda_{ij} I_j^0 \frac{d \log W_j}{dt_{i\ell}} = 0. \tag{5}$$

These total derivatives run through the competitive equilibrium and include wage, expenditure, and tariff-revenue responses. Partial derivatives at fixed wages would omit the terms-of-trade effects of a tariff, which are the source of both the domestic gain and the foreign loss. Appendix A.3 derives them from the implicit-function theorem, and the same derivatives are the gradients on which the policy game is solved in every regime. Define the margins in domestic-income units as $y_{i\ell} = d \log W_i / dt_{i\ell}$ and $x_{i\ell j} = -(I_j^0 / I_i^0)\, d \log W_j / dt_{i\ell}$. The condition is $y_{i\ell} = \sum_{j \neq i} \lambda_{ij} x_{i\ell j}$. At an interior choice, marginal domestic gains balance weighted foreign losses.

With three sectors, each bilateral weight has at most three strongly informative direct tariff observations. Identification from other cells relies on smaller general-equilibrium

effects. The baseline therefore imposes a common weight $\lambda_{ij} = \lambda$ and estimates

$$y_{i\ell} = \lambda X_{i\ell} + \varepsilon_{i\ell}, \qquad X_{i\ell} = \sum_{j \neq i} x_{i\ell j}, \qquad \ell \in \mathcal{E}_i, \tag{6}$$

where $\mathcal{E}_i$ is government i's interior sample. Least squares without an intercept pools these normalized equalities. The coefficient is identified by cross-cell variation in the foreign welfare margins. This variation is related to the market-power channel in Broda et al. (2008) and Ludema and Mayda (2013). Standard errors are clustered by importer, the level at which tariffs are chosen.

The residual includes omitted nontariff policies, partner-specific preferences, measurement error, and aggregation. Interpreting the coefficient as a welfare weight requires

$$\mathbb{E}[X_{i\ell}\varepsilon_{i\ell} \mid \ell \in \mathcal{E}_i] = 0. \tag{7}$$

Under a correctly specified common-weight model, informative interior samples satisfying this condition identify the same parameter. The condition can fail through simultaneity, since the foreign margin is evaluated at observed tariffs that the residual also shapes, and through selection into the interior sample, as when protectionist motives push tariffs to their ceilings. A free-trade instrument addresses the first, since a margin evaluated at zero tariffs does not depend on the residual, but not the second, and its own exclusion restriction is likewise untestable. The estimate is therefore consistent under the stated condition rather than verified.

3.3 Estimates and interpretation

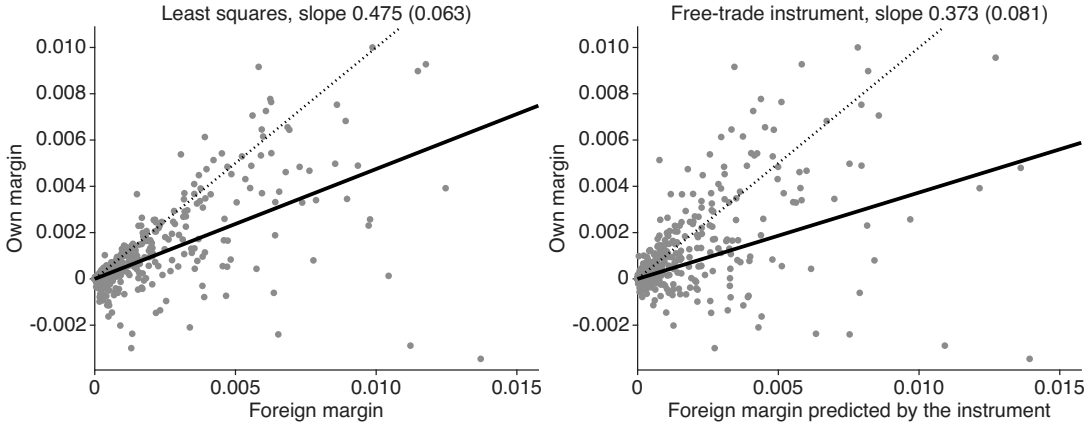
Table 2 reports a pooled interior weight $\widehat{\lambda}$ of 0.47, with a standard error of 0.06 clustered on the 27 governments that contribute interior cells, and the common weight applies to all 32. In the objective (1), this assigns approximately half as much value to a dollar of foreign real income as to a domestic dollar. Figure 1 plots the margins for the 961 cells with the least-squares and instrumental-variable fits. Both fitted slopes lie between zero and full internalization. The remaining checks ask whether this estimate is sound and whether it is the weight to carry into the counterfactuals, against the higher estimate from all cells and the lower estimates from tighter samples.

Table 2: Estimates of the foreign welfare weight

	$\widehat{\lambda}$	s.e.	Cells
Main estimates			
Baseline, interior cells	0.47	0.06	961
Free-trade instrument	0.37	0.08	961
All nondegenerate cells	0.83	0.05	2,498
Corner cells, interior coverage, partner groups, and fixed effects			
Interior plus zero-tariff cells	0.56	0.05	1,096
Interior plus nonpreferential ceiling cells	0.57	0.05	1,437
Interior coverage of at least one quarter	0.27	0.07	607
Interior coverage of at least one half	0.21	0.09	484
Weighted by interior coverage	0.31	0.07	961
Advanced partners	0.53	0.08	961
Developing partners	0.46	0.07	961
Importer fixed effects added	0.45	0.07	961

Notes: No-intercept least-squares regressions of the domestic welfare margin on the foreign margin, both divided by government baseline income, with importer-clustered standard errors. The instrument is the foreign margin at free trade, and its first-stage F statistic uses the same clustering. The nondegenerate sample adds 1,402 ceiling-class and 135 positive-width zero-tariff cells to the interior cells and excludes 402 singleton domains. The corner-cell rows add the zero-tariff cells and the 476 ceiling-class cells of nonpreferential partners to the interior sample as equalities. Interior coverage is the baseline-import share on products strictly between zero and their bindings. The partner rows are two slopes from one regression. The weight column of Table E2 reports the estimate under further specifications.

Figure 1: Domestic gains and foreign losses from marginal tariff increases



Notes: Each point is an interior government-partner-sector cell, with both margins divided by government baseline income. The vertical axis is the domestic welfare margin, and the horizontal axis is the foreign margin in the left panel and its first-stage prediction from the free-trade instrument in the right panel. Solid lines are the least-squares and instrumental-variable fits of Table 2, and dotted lines show full internalization, $\lambda = 1$. Display limits at the first and ninety-ninth percentiles omit 36 and 35 cells.

The instrumental-variable estimate is 0.37, and each estimate lies within the other's confidence interval, so the comparison gives no evidence of simultaneity bias. The baseline retains least squares, which evaluates the estimating condition at the observed tariffs, since the instrument's exclusion restriction is no more testable than the least-squares condition. Dyadic clustering and Anderson-Rubin sets also exclude zero and full internalization, and Appendix C.4 reports them with the sensitivity to bounded violations of the estimating moments.

The broad estimate of 0.83 counts restraint that the ceilings enforce. Applying the interior equality conditions to all 2,498 nondegenerate cells treats a tariff at a binding ceiling as an unconstrained optimum, although the government would raise it if it could, and so attributes the ceiling's restraint to cooperation. Table 2 shows which cells carry the difference. Adding the 135 zero-tariff cells to the interior sample gives 0.56, whether they enter as equalities or as inequalities.[4] Adding the 476 ceiling-class cells of nonpreferential partners, MFN tariffs at or within five points of their bindings, gives 0.57. The remaining 926 cells of the broad sample are preferential rates, which sit at their ceilings by construction, so the gap between 0.57 and 0.83 is preferential rates read as cooperation. The counterfactuals impose those ceilings explicitly, so a weight that already contained their restraint would count it twice, and the baseline keeps 0.47. The broad specification is closest to Adão et al. (2026), whose estimated weights summarize institutional restraint and average 0.75 in 2001 and approximately 0.88 in 2019.

Interior coverage is the share of a cell's imports on products free to move in either direction. Cells with interior coverage of at least one half give a weight of 0.21, and weighting cells by interior coverage gives 0.31 (Table 2). Products at bindings push the estimate up and products at zero push it down, so the decline is consistent with binding restraint being read as cooperation without establishing it. High-coverage cells also differ in partners and sectors, so the decline cannot be attributed to aggregation (Appendix C.4). The baseline therefore keeps the full interior sample, which uses every cell free to move on average and the widest set of governments. What this cannot rule out is that bindings on some products within interior cells still restrain them, in which case 0.47 overstates cooperation. Appendix Table E6 reports the value and its attribution at lower weights.

[4] As equalities, the zero cells enter the least-squares regression as if the zero tariff satisfied the interior first-order condition $y_{i\ell} = \lambda X_{i\ell}$. As inequalities, they enter a censored likelihood in which $y_{i\ell} \leq \lambda X_{i\ell}$ up to the residual, so that the government would not gain from a marginal tariff increase from zero (Appendix C.4).

Governments weight advanced partners somewhat more than developing ones, 0.53 against 0.46 (Table 2), consistent with greater restraint toward partners better able to retaliate, although the difference is statistically imprecise. The estimate does not depend on any single importer, since leaving out one government at a time moves it only between 0.44 and 0.51, and importer fixed effects give 0.45. It is more sensitive to partners, ranging from 0.36 without the rest of world to 0.66 without China as an exporter (Appendix C.4), which indicates a lower weight on China than on other partners.

The cells outside the sample provide an out-of-sample test of the weight. At a ceiling, optimality under the estimated weight requires $y_{i\ell} - \widehat{\lambda} X_{i\ell} \geq 0$, a desired tariff above the ceiling, and at zero the reverse. The ceiling condition holds on 90 percent of imports at ceiling contacts, so ceilings restrain desired increases on most of this trade. The zero condition holds on almost none, so the common weight does not rationalize zero tariffs, which omitted motives, measurement error, or aggregation may explain.[5] These cells carry 1.4 percent of active imports, and reading them as choices moves the weight only to 0.56 (Table 2), so the misfit is small in trade and does not move the estimate outside its interval.

4 The Baseline Value of the System

This section quantifies the welfare gains protected by the trading system in the 2017 baseline, before any productivity change.[6] It measures the aggregate value of the system, separates the contributions of cooperation and commitments, which are substitutes in the baseline, and examines how the value is distributed across economies. These baseline results are the reference point for the productivity shifts in Section 5.

4.1 Cooperation and commitments as substitutes

At the baseline weight and ceiling specification, each restraint protects most of the system's aggregate gain. Table 3 reports a world value of 0.81 percent of real income relative to

[5] At zero, the condition asks whether a government gains from a small initial tariff after counting the weighted partner loss, and any importer with market power does, so the estimated weight implies a positive desired tariff almost everywhere. Half of the trade in these cells is extraction, whose zero rates lower input costs that the labor-only model cannot value, and most of the rest is imported by Singapore and Switzerland, whose near-zero schedules reflect a unilateral stance rather than a bilateral calculation.

[6] Each regime is solved by projected Newton iteration on the first-order and bound conditions, with the competitive equilibrium and its derivatives recomputed at every step (Appendix B.3).

Table 3: The baseline value of the trading system

	World	Median
Solved policy equilibria		
Commitments only	0.74	1.75
Cooperation only	0.76	1.27
Full system	0.81	1.93
Reference schedules		
Observed 2017 tariffs	0.55	1.54
Free trade	0.84	2.10

Notes: Entries are log percentage points of real income relative to the noncooperative equilibrium. The first panel reports the regimes defined in Table 1. The second panel values two schedules that are not equilibria of the policy game. Appendix Table E3 reports country values.

noncooperation and a median value of 1.9 percent.[7] Commitments alone deliver 0.74 percent and cooperation alone 0.76 percent, or 92 and 94 percent of the full gain. The commitment value does not depend on the estimated weight, since both regimes in its definition carry zero weights.

The two restraints are substitutes in the baseline, since their separate gains sum to more than their combined gain for every optimizing government. Either restraint alone removes most of the noncooperative tariffs. Commitments alone cut advanced-economy mean tariffs from about 16 percent under noncooperation to 2.6 percent in manufacturing and 1.0 percent in extraction, and cooperation alone protects a similar value. The loss from a tariff is roughly proportional to its square, so whichever restraint is added second has little loss left to remove.

The overlap is also visible in which ceilings bind. With commitments alone, ceilings bind on 1,245 choices covering 82.3 percent of baseline imports on positive-width institutional domains. In the full system, they bind on 916 choices covering 68.5 percent. Cooperation moves about a quarter of the commitment-only contacts below their ceilings, and a contact records only that a ceiling binds, not by how much.

The full system uses tariffs solved under the estimated weight and ceilings. They differ from observed 2017 tariffs because the objective fits observed choices only on average and omits sector-specific protection motives. The solved system liberalizes agriculture more

[7] Reported values are $100v_i$, the log gain in (3) multiplied by 100, in log percentage points. A value of one corresponds to approximately one percent higher real income, so the text rounds to percent. After a productivity shock, $100\Delta v_i$ is the change in this value. World values weight the 34 named economies by fixed baseline income, and medians are across the 32 optimizing governments. Income covers the three modeled tradable sectors and includes tariff revenue. Appendix B.2 defines a complementary measure summing national gains at fixed baseline prices.

than observed policy does, with advanced-economy mean tariffs of 1.9 rather than 8.2 percent (Appendix Table E4). The observed schedule yields a world value of 0.55 percent and a median value of 1.5 percent, compared with 0.81 and 1.9 under the solved system.

The system's aggregate gain is close to the free-trade benchmark. Free trade yields 0.84 percent for the world and 2.1 percent for the median economy, against 0.81 and 1.9 for the system. These are benchmark comparisons, not welfare bounds.

The loss from a trade war is smaller here than in related work. The median gain from the observed schedule, 1.5 percent, compares with 2.2 percent in Adão et al. (2026) and an average loss of 2.9 percent in Ossa (2014). Noncooperative tariffs are low here because the only protection motive in the model is the terms of trade, which is relatively weak with high trade elasticities and 32 separate governments of limited market power. Three sectors also average away product-level differences in trade exposure that a product-level Nash equilibrium exploits.

4.2 Incidence across economies

All 32 optimizing governments gain from the full system. The gains range from 0.14 percent for China to 6.3 percent for Mexico (Appendix Table E3). The largest economies have relatively small gains, 0.5 percent for the United States and Japan, 0.4 for the EU, and 0.3 for India. A large domestic market limits exposure to partners' tariffs, and own tariffs recover some income through the terms of trade, the logic of Johnson (1953).

The restraints' relative value is associated with economy size, but size alone does not determine it. Cooperation alone is worth more than commitments alone for eight governments: Argentina, China, India, Japan, Pakistan, Turkey, the United States, and the EU. Commitments alone are worth more for the other 24. Across all 32 governments, median values are 1.75 percent under commitments and 1.27 under cooperation (Appendix Table E5).

Two features of the model can account for this pattern. The weight applies to partners' real income in levels at baseline prices rather than in proportional terms, so a tariff that costs a large partner more income is restrained more, and cooperation protects large economies most. Ceilings apply independently of those welfare weights, through bound MFN rates and preferential rates that are often zero. They provide smaller economies with market access beyond that delivered by cooperation alone. In the baseline, these

economies generally gain more from that access than they lose by binding their own tariffs.

Every government prefers the system to the war, but every one would also gain from discarding both restraints alone while partners keep their system tariffs, by a median of 0.77 percent for advanced economies and 0.39 for developing economies (Appendix B.3). Preferring the system to the war is therefore necessary but not sufficient for it to sustain itself. Whether the threat of the war deters unilateral deviation depends on discounting and retaliation, which lie outside the model.

5 Productivity Shocks and the Value of the System

This section quantifies how productivity changes alter the value of the trading system. Section 2 showed that only relative productivity changes move the value, through exposure at fixed tariffs and the tariff response of (4), with no sign prediction in a multi-country and multi-sector setting. Welfare weights, ceilings, and all other primitives stay fixed, the estimation residual is not carried into the counterfactuals, and the policy equilibria are re-solved after each shock.

The three counterfactual analyses are chosen to span the forms a shift in comparative advantage can take. The first widens and narrows comparative advantage by shifting productivity in opposite directions in each economy's strongest and weakest sectors, a symmetric shift across every economy. The second concentrates the change in one large economy in each direction, a contraction of Chinese manufacturing and a shale-motivated expansion of United States extraction. The third applies projected climate-related productivity changes of mixed sign across most economies and sectors. Table 4 compares the five shifts by the change in the system's value with its exposure and policy terms (first panel) and by the change in the value of each restraint alone (second panel).

5.1 Widening and narrowing comparative advantage

The first counterfactual raises productivity by about 4.6 percent in each economy's most competitive sector and lowers it by the same log amount in its least competitive one. The reverse shift moves the two sectors toward each other and can reverse their ranking when the initial gap is small. The shift is comparable to two decades of the within-country convergence documented by Levchenko and Zhang (2016) at the median gap between an

Table 4: Productivity shocks and changes in the value of the system

Value of the system	World			Median
	Total	Exposure	Policy	Total
Widening comparative advantage	2.2	6.8	−4.6	11.5
Narrowing comparative advantage	−0.1	−2.5	2.4	−7.4
Chinese manufacturing contraction	6.7	8.3	−1.6	−8.1
United States extraction expansion	3.1	9.7	−6.6	−4.3
Climate change	−0.8	−1.0	0.3	−1.3

Value of each restraint alone	Commitments alone		Cooperation alone	
	World	Median	World	Median
Widening comparative advantage	2.5	10.9	0.8	−0.8
Narrowing comparative advantage	0.1	−7.4	−0.7	−0.8
Chinese manufacturing contraction	7.3	−7.7	2.0	−1.2
United States extraction expansion	3.1	−5.1	1.3	0.4
Climate change	−0.8	−0.7	−0.4	−1.5

Notes: Percent changes in the value of the trading system (first panel) and in the value of commitments alone and of cooperation alone (second panel), each relative to its own baseline, 0.81, 0.74, and 0.76 log percentage points for the world. Exposure is the first term of (4) and policy the sum of its two tariff-response terms, computed in the finite form (B.2). Medians are across the 32 governments of each economy's change relative to its own baseline value. Every regime, including cooperation alone, is re-solved after each shock. Rounding may prevent exact addition. The Chinese contraction lowers manufacturing productivity by 35 percent and the United States expansion raises extraction productivity by 27 percent, the former reversing 1995 to 2007 growth and the latter matching mining productivity growth over 2007 to 2017. Appendix Table F1 reports point values for the productivity experiments.

economy's sectors. Appendix D.1 gives the calibration details of the shock.

Widening raises the value of the trading system by 2.2 percent for the world and by 11.5 percent for the median economy. The tariff response runs against two-country intuition. In the two-country Ricardian model of Opp (2010), wider comparative advantage raises noncooperative tariffs, whereas here they fall, from 16.2 to 15.3 percent for advanced economies and from 15.7 to 14.4 for developing economies, and system ceiling contacts fall from 916 to 875. Narrowing reverses both movements. With multiple heterogeneous countries, a wider advantage also shifts market power and diverts trade across partners, which can reverse the two-country response.

The two channels work against each other. At each regime's baseline tariffs, widening generates larger welfare gains under the system than under noncooperation, so exposure alone would have raised the value by 6.8 percent. Tariff adjustment offsets about two thirds of that increase, since noncooperative tariffs fall under widening and the war the

system averts becomes less costly. Under narrowing, a negative exposure contribution of 2.5 percent is almost offset by a positive policy contribution of 2.4 percent.

5.2 Concentrated shocks in China and the United States

The second counterfactual analysis concentrates the shift in one large economy at a time, with opposite signs. The first reduces Chinese manufacturing productivity by 35 percent, reversing the 3.5 to 3.7 percent annual growth that Hsieh and Ossa (2016) estimate for 1995 to 2007. The second raises United States extraction productivity by 27 percent, the measured growth of mining total factor productivity over 2007 to 2017 in the Bureau of Labor Statistics accounts, with the broad extraction sector standing in for shale and the shock applied to a baseline that already contains that growth (Appendix D.2).

Both shocks raise the world value of the trading system, by 6.7 percent under the Chinese contraction and 3.1 percent under the United States expansion, and both lower the median economy's value, by 8.1 and 4.3 percent. In each case the gain comes from the shocked economy, whose large income weight makes its increase outweigh the net decline for its partners. China's value more than triples, from 0.14 to 0.49 log percentage points. Its real income falls, but low tariffs let imports replace the lost manufacturing output, whereas noncooperative tariffs would leave it dependent on its own manufacturing, and this import channel outweighs its shrinking exports. The United States' value more than doubles, from 0.54 to 1.24. Its expanded extraction output needs export markets that noncooperative tariffs would nearly close, given the sector's high trade elasticity, and the expansion draws in more manufacturing imports. The shocked economy relies more on trade that a war would tax in both cases, imports for China and export outlets for the United States, so the sign of the productivity change alone does not determine the sign of the aggregate response.

The value falls for partners on net under both shocks (Appendix Table E5). Under the Chinese contraction it rises for manufacturing exporters that replace China, including Japan, South Korea, Taiwan, and Thailand, and falls for commodity exporters facing lower Chinese demand, most sharply for Australia, and for the United States as its imports from China decline. Under the United States expansion it rises for extraction importers such as India, the EU, Japan, and South Korea, and falls for competing extraction exporters, including Canada, Norway, Australia, and Kazakhstan. Exposure alone would have raised

the world value by 8.3 and 9.7 percent, and tariff responses offset a fifth of the first and two thirds of the second (Table 4). Less trade passes through cells at their ceilings once China's exports contract, yet the exposure gain still exceeds the policy offset, so the volume of protected trade alone does not determine what the system is worth.

Larger expansions produce different tariff responses. At 50 and 80 percent, the value of the trading system rises by 12.7 and 35.4 percent (Appendix Table F1). At 80 percent, mean noncooperative tariffs among advanced importers rise from 16.2 to 17.2 percent, and tariff responses reinforce the exposure term. The response is therefore more than proportional to the shock, and at the largest magnitude the tariff response changes sign.

5.3 Climate change

The climate scenario applies projected productivity changes from Nath (2025) to agriculture and manufacturing in most economies, with extraction unchanged. Appendix D.3 provides country-sector details on the shocks. With observed tariffs fixed, the shock reduces world income by approximately 2.3 percent, and the largest single-economy loss is about 7 percent. The world loss is the same to two decimals in every solved regime, so the system neither cushions nor amplifies it.

Climate change has only a small effect on the value of the trading system. The world value falls by 0.8 percent and the median economy's value by 1.3 percent, with exposure contributing −1.0 percent and tariff responses 0.3 percent (Table 4). The shock lands where the system protects little. Agricultural productivity changes range from −24 to 20 percent across economies, but agriculture carries only about 3.5 percent of modeled trade. Manufacturing, where the trade at binding ceilings is concentrated, changes by −4 to below 1 percent. The agricultural trade that ceilings do protect is also little affected, since the economies with the most protected agricultural markets, Japan, South Korea, Switzerland, and Norway (Anderson et al., 2013), are among those whose agricultural productivity changes least or rises. Further, the common component of the shock, about −2.4 percent under expenditure weights, leaves proportional value unchanged by Corollary 1, so only the relative changes that remain can move it.

Climate change redistributes the system's gains across economies while the total is nearly unchanged. For individual economies, the value of the system changes by between about −12 percent for Pakistan and 12 percent for Argentina, and it falls for 19 of

the 32 governments (Appendix Table E3). The incidence follows the mechanism of the concentrated shocks. Value rises mainly for economies whose manufacturing productivity falls, such as China, Turkey, Ukraine, and Taiwan, consistent with greater reliance on the manufacturing imports the system protects. It falls for exporters whose losses are concentrated in agriculture, such as Australia, South Africa, and Pakistan, whose shrinking exports are in a sector the system protects little. Argentina and Thailand gain despite purely agricultural damage. Australia's decline is mainly exposure, Argentina's increase reflects exposure partly offset by policy adjustment, and Pakistan's decline combines both channels.

These results complement research on adaptation through trade and reallocation (Gouel and Laborde, 2021; Nath, 2025). They show that the trading system protects no more value under climate change than in the baseline. The shock falls on agriculture, where ceilings protect little trade, so tighter agricultural ceilings could in principle raise the value the system protects under climate change. Evaluating such ceilings is left for future research.

5.4 The responses of the two restraints

The two restraints respond very differently in these analyses. The value of commitments alone changes almost as much as the value of the system, by 7.3 percent against 6.7 under the Chinese contraction, whereas in all shifts but narrowing the value of cooperation alone changes in the same direction by between a quarter and a half as much. The two stand-alone values overlap, so their changes are not additive and commitments alone can rise slightly more than the system. Ceilings are fixed caps, so when a shift raises the tariff a government would like to set in a cell where the ceiling binds, the gap between the capped and the noncooperative tariff rises one for one. To a first approximation, cooperation lowers tariffs in proportion to the weight on partner losses, so the same shift raises the gap between cooperative and noncooperative tariffs only in proportion to the weight, about one half. Under narrowing, cooperation alone moves more than commitments alone, although both changes are below one percent. The medians show the same contrast across economies. The value of commitments is strongly redistributed, with median changes between -8 and 11 percent, while the value of cooperation barely moves for the median economy.

Where a shift lands matters more than its size. The two concentrated shocks move the world value by 3 to 7 percent, whereas the climate shock, which lowers world income by 2.3 percent but falls mainly on agriculture, leaves it almost unchanged. What comparative advantage does to the value of the system therefore depends on whether it moves the trade the ceilings protect, and the commitments carry most of that movement.

6 Sensitivity and Scope

The checks vary the estimation, the policy assumptions, the shock construction, and the aggregation, with the variants listed in Appendix Table E2. The baseline value and its attribution are the results most sensitive to the weight and the ceilings. Cooperation alone recovers 94 percent of the full-system value at the baseline weight, 85 percent at the instrumental-variable estimate, and 68 percent at an illustrative weight of 0.26 below the confidence interval (Appendix Table E6). Scaling the trade elasticities by 0.8 and 1.2 moves the baseline value to 0.995 and 0.684 log percentage points.

The preferential ceilings are the part of the institutions that may change, since agreements are signed and revised more often than bound rates. The baseline weight of 0.47 is estimated with the preferential ceilings in place. Removing them from the counterfactual policy domains while keeping that weight, so that only the WTO bindings constrain tariffs, lowers the baseline value from 0.810 to 0.769 log percentage points, a decline of 5 percent, and the climate response stays small (Appendix Table E1). Re-estimating the weight without the preferential ceilings gives 0.837 (standard error 0.09), since restraint that those ceilings absorbed is then read as cooperation, and the value at that weight is 0.836. The value the system protects therefore does not depend on the preferential ceilings.

The signs of the counterfactual shifts survive alternative constructions. The revealed-specialization ranking and rank-preserving proportional shocks retain the widening and narrowing signs, the concentrated shocks keep their signs at other magnitudes, and all five keep their signs at both endpoints of the weight interval (Appendix Tables E7 and F1). Reversing the Chinese shock also raises aggregate value, whereas reversing the extraction shock lowers it (Appendix F).

The climate response stays small in every variant. World value changes by -1.2 to 1.2

percent across the seventeen estimation and policy variants, the separate agricultural and manufacturing shocks, and the alternative agricultural damage projections, although the sign turns positive under one of the three projections (Appendix Table E2).

Aggregation matters for the concentrated shocks. With equal weights across the 34 named economies, average value falls by 12.6 percent under the Chinese contraction and 10.2 percent under the extraction expansion, against their positive income-weighted responses, and the climate decline is 1.6 percent (Appendix Table E7).

7 Conclusion

This paper measures the value of tariff restraint and its response to productivity shifts, separating cooperation from commitments. The foreign welfare weight is about one half. At this weight and the baseline ceilings, the system is worth 0.81 percent of world income and 1.9 percent for the median economy relative to noncooperation. Each restraint alone delivers most of the aggregate value. Larger economies tend to value cooperation more relative to commitments, and the reverse holds for smaller economies.

A shock's sign alone does not determine its effect on the system's value. In these analyses, widening comparative advantage raises aggregate value and narrowing lowers it slightly. The Chinese manufacturing contraction and United States extraction expansion both raise the world value while lowering it for most partners, since each shocked economy comes to rely more on trade that noncooperative tariffs would tax. Across these shifts, commitments carry most of the response, so restraints that substitute for each other in the baseline do not substitute in their responses. In these shifts, the system's resilience therefore rests mainly on its bindings. Climate change has a small aggregate effect and redistributes value toward economies whose manufacturing productivity falls and away from those whose losses fall on agriculture.

Three extensions follow. Productivity changes could be allowed to affect enforcement and renegotiation (Beshkar and Bond, 2017), the value of reduced policy uncertainty could be included (Handley, 2014; Limão and Maggi, 2015), and export restrictions could be allowed in response to agricultural shocks (Martin and Anderson, 2012; Hsiao et al., 2026).

Three limitations qualify the results. First, the common weight does not rationalize

observed agricultural protection or zero applied tariffs below positive bindings, so the solved system does not reproduce the observed schedule, and the results describe the estimated institutions rather than the observed tariffs. Second, the division of value between cooperation and commitments depends on the ceiling specification and on the weight, which is identified from the 14 percent of active imports on interior tariffs. Third, world values weight economies by tradable gross output rather than GDP, so a different weighting would change the aggregates.

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Appendix

Appendix A provides the competitive equilibrium and its representation in relative changes. Appendix B covers the policy game, its finite decomposition, and its computation, Appendix C documents the data and the calibration, and Appendix D the productivity scenarios. Appendix E reports per-economy results and further tables, and Appendix F reports the additional productivity shifts.

A Competitive equilibrium and hat algebra

The competitive equilibrium is the standard multi-sector model of Eaton and Kortum (2002) with tariff revenue rebated, as in Caliendo and Parro (2015) without intermediate inputs. This appendix fixes the notation used in the derivations and the estimation, writes the equilibrium in relative changes following Dekle et al. (2008), and states the regularity conditions for welfare derivatives.

A.1 The competitive equilibrium

The countries and sectors, the expenditure shares α_{is}, the technology parameters T_{js} and θ_s, the tariffs t^s_{ji}, expenditure I_i, the price index P_i, and real income W_i are those of Section 2.1. Country i has labor $L_i > 0$, mobile across its sectors and immobile internationally, so every active sector pays the wage w_i, and production uses labor alone under constant returns. The expenditure shares are positive and sum to one, preferences within each sector are CES with elasticity $\rho_s > 1$ over a unit continuum of varieties, and with P_{is} the sectoral CES price index, sectoral expenditure is $\alpha_{is} I_i$ and $P_i = \prod_s (P_{is}/\alpha_{is})^{\alpha_{is}}$. Each variety's labor productivity in country j and sector s is drawn independently from the Fréchet distribution with scale $T_{js} > 0$ and shape $\theta_s > \rho_s - 1$, so that multiplying T_{js} by q^{θ_s} for any $q > 0$ scales every draw by q. Delivery from j to i requires $d^s_{ji} \geq 1$ units per unit delivered, with $d^s_{ii} = 1$, and $t^s_{ii} = 0$. Define

$$\tau^s_{ji} = d^s_{ji}(1 + t^s_{ji}), \qquad \Phi_{is} = \sum_j T_{js}(w_j \tau^s_{ji})^{-\theta_s}. \tag{A.1}$$

Sourcing each variety from the cheapest supplier gives the expenditure shares and sectoral price indices

$$\pi_{ji}^{s} = \frac{T_{js}(w_j \tau_{ji}^{s})^{-\theta_s}}{\Phi_{is}}, \tag{A.2}$$
$$P_{is} = A_s \Phi_{is}^{-1/\theta_s}, \qquad A_s = \left[\Gamma\left(1 + \frac{1-\rho_s}{\theta_s}\right)\right]^{1/(1-\rho_s)},$$

where π_{ji}^{s} is country j's share of country i's sectoral expenditure. Let M_{ji}^{s} denote tariff-exclusive sales from j to i. Producer receipts and tariff revenue are

$$M_{ji}^{s} = \frac{\alpha_{is} I_i \pi_{ji}^{s}}{1 + t_{ji}^{s}}, \qquad R_i = \sum_{j,s} t_{ji}^{s} M_{ji}^{s}. \tag{A.3}$$

where receipts include payment for the labor embodied in units lost in transit, which accrues at origin. Tariff revenue is rebated to households, and there are no international transfers, so

$$I_i = w_i L_i + R_i,$$
$$w_j L_j = \sum_{i,s} M_{ji}^{s}. \tag{A.4}$$

Sectoral employment follows from sales, $L_{js} = \sum_i M_{ji}^{s}/w_j$, and combining these conditions gives balanced trade at tariff-exclusive values. A competitive equilibrium for $(\mathbf{T}, \mathbf{t})$ is a vector of positive wages and expenditures satisfying (A.2) and (A.3)–(A.4), with one wage normalized to one. Institutional regimes affect the equilibrium only through tariffs.

A.2 Equilibrium in relative changes

With superscript 0 marking the baseline as in Section 2.1, write $\widehat{x} = x/x^0$ for a strictly positive variable. Expenditure weights, labor endowments, and iceberg costs are fixed, and baseline tariffs need not form a policy equilibrium in any regime. Define

$$\widehat{\tau}_{ji}^{s} = \frac{1 + t_{ji}^{s}}{1 + t_{ji}^{s,0}}, \qquad \widehat{\Phi}_{is} = \frac{\Phi_{is}}{\Phi_{is}^{0}} = \sum_j \pi_{ji}^{s,0} \widehat{T}_{js} (\widehat{w}_j \widehat{\tau}_{ji}^{s})^{-\theta_s}. \tag{A.5}$$

Then shares and prices change as

$$\widehat{\pi}^s_{ji} = \frac{\widehat{T}_{js}(\widehat{w}_j\widehat{\tau}^s_{ji})^{-\theta_s}}{\widehat{\Phi}_{is}},$$
$$\widehat{P}_{is} = \widehat{\Phi}_{is}^{-1/\theta_s}, \qquad \widehat{P}_i = \prod_s \widehat{P}_{is}^{\alpha_{is}}.$$

for simultaneous changes in all technologies and tariffs. Tariff revenue as a share of expenditure is

$$r_i = \frac{R_i}{I_i} = \sum_s \alpha_{is} \sum_j \pi^s_{ji} \frac{t^s_{ji}}{1+t^s_{ji}},$$

with $0 \le r_i < 1$ for finite nonnegative tariffs, so the household budget gives

$$I_i = \frac{w_i L_i}{1-r_i}, \qquad \widehat{I}_i = \widehat{w}_i \frac{1-r^0_i}{1-r_i},$$

which is well defined at zero baseline revenue. With the baseline producer-sales weights

$$\omega^{s,0}_{ji} = \frac{M^{s,0}_{ji}}{w^0_j L_j}, \qquad \sum_{i,s} \omega^{s,0}_{ji} = 1,$$

receipts and labor-market clearing become

$$\widehat{M}^s_{ji} = \widehat{\pi}^s_{ji}\widehat{I}_i \frac{1+t^{s,0}_{ji}}{1+t^s_{ji}},$$
$$\widehat{w}_j = \sum_{i,s} \omega^{s,0}_{ji} \widehat{M}^s_{ji}. \tag{A.6}$$

Equations (A.5)–(A.6), with one $\widehat{w}_j = 1$, determine the equilibrium in relative changes on the chosen branch, and welfare satisfies

$$\widehat{W}_i = \frac{\widehat{I}_i}{\widehat{P}_i} = \widehat{w}_i \frac{1-r^0_i}{1-r_i} \prod_s \widehat{\Phi}_{is}^{\alpha_{is}/\theta_s}.$$

The system requires only baseline bilateral shares, producer-sales weights, expenditure weights, tariffs, and trade elasticities. Cells with zero baseline trade remain at zero, corresponding to prohibitive trade costs; trade-flow hats apply only to positive baseline cells. At fixed baseline prices the government objective becomes

$$P^0_i W_i + \sum_{j \ne i} \lambda_{ij} P^0_j W_j = I^0_i \widehat{W}_i + \sum_{j \ne i} \lambda_{ij} I^0_j \widehat{W}_j,$$

so baseline expenditures and the cooperation weights suffice to aggregate welfare changes in the policy objective.

A.3 Equilibrium derivatives

This subsection applies the standard implicit-function calculation to the equilibrium system. The policy game evaluates welfare after markets clear. Normalize $w_N = 1$ and collect the remaining log wages and all log expenditures in

$$\boldsymbol{\xi} = (\log w_1, \ldots, \log w_{N-1}, \log I_1, \ldots, \log I_N).$$

Let $\boldsymbol{\mu}$ collect log technologies and tariff levels. Substituting sourcing shares into the expenditure and labor-market equations gives the square system

$$\mathbf{F}(\boldsymbol{\xi}, \boldsymbol{\mu}) = \mathbf{0}.$$

It contains the N expenditure equations $(1 - r_i)I_i - w_i L_i = 0$ and the $N - 1$ labor-market equations $w_j L_j - \sum_{i,s} M_{ji}^s = 0$, $j < N$. The omitted labor-market equation follows from their sum and the tariff-revenue identity.

If $\mathbf{F}$ is continuously differentiable and $\mathbf{F}_{\boldsymbol{\xi}}$ is nonsingular, the implicit-function theorem gives a locally unique, differentiable equilibrium branch. Along that branch,

$$\frac{\partial \boldsymbol{\xi}}{\partial \mu_k} = -\mathbf{F}_{\boldsymbol{\xi}}^{-1}\mathbf{F}_{\mu_k}, \qquad \frac{d \log W_i}{d \mu_k} = \frac{\partial \log W_i}{\partial \mu_k} - \frac{\partial \log W_i}{\partial \boldsymbol{\xi}}\mathbf{F}_{\boldsymbol{\xi}}^{-1}\mathbf{F}_{\mu_k}. \tag{A.7}$$

Partial derivatives on the right hold $\boldsymbol{\xi}$ fixed. The total welfare derivative includes changes in wages, expenditures, and tariff revenue. Government tariff incentives use this total derivative. Imposing equal wages during a unilateral deviation from symmetry would generally omit the terms-of-trade response.

Nonsingularity does not establish global uniqueness of the competitive or policy equilibrium. Differentiating a policy equilibrium also requires regularity of the policy system.

A.4 A common productivity scale

Corollary 1 (Common productivity scale)**.** *Fix government welfare weights, baseline prices, policy domains, and all nontechnology primitives. For $q > 0$, let $T_{js}(q) = q^{\theta_s} T_{js}$ for every country and sector. Under corresponding equilibrium selections, tariff profiles are unchanged in each regime and $W_i^R(\mathbf{T}(q)) = qW_i^R(\mathbf{T})$, so v_i, v_i^{com}, and v_i^{coop} are unchanged.*

The corollary is a homogeneity property, not a new result, and the proof below restates the standard argument and checks the one step that the present environment changes. The competitive-equilibrium part is standard in Ricardian models, since sourcing shares depend on technologies only relative to one another (Eaton and Kortum, 2002). The invariance of optimal and noncooperative tariffs to absolute productivity is likewise known, since Opp (2010) shows that the uniform optimal tariff in the continuum Ricardian model depends on relative productivity-adjusted size and Felbermayr et al. (2013) obtain the same result in a one-sector model with monopolistic competition and heterogeneous firms. The remaining step is to check that the tariff domains and welfare objective preserve this scaling.

Fix technologies $\mathbf{T}$ and a scalar $q > 0$, and define

$$T_{js}(q) = q^{\theta_s} T_{js} \quad \text{for every } j, s. \tag{A.8}$$

The standard argument runs as follows. Substituting (A.8) into (A.1) multiplies each Φ_{is} by q^{θ_s}, so at fixed tariffs the sourcing shares, normalized wages, expenditures, tariff revenue, and sectoral employment are unchanged, sectoral price indices are divided by q, and, since expenditure weights sum to one,

$$P_{is}(q) = q^{-1} P_{is}(1), \qquad P_i(q) = q^{-1} P_i(1), \qquad W_i(q) = qW_i(1).$$

The step to check is the policy game. Ceilings and policy domains are constraints in tariff space, which the scaling does not touch, and each government's objective is linear in real incomes valued at fixed baseline prices, so it is multiplied by q at every tariff profile. Best responses and policy-equilibrium sets are therefore unchanged in every regime, and

corresponding equilibrium selections satisfy

$$W_i^{\text{sys}}(q) - W_i^{\text{nc}}(q) = q[W_i^{\text{sys}}(1) - W_i^{\text{nc}}(1)], \qquad \log \frac{W_i^{\text{sys}}(q)}{W_i^{\text{nc}}(q)} = \log \frac{W_i^{\text{sys}}(1)}{W_i^{\text{nc}}(1)}.$$

The same holds for v_i^{com} and v_i^{coop}, each of which is the log ratio of welfare in two regimes. Every proportional value is unchanged and every gain measured at fixed baseline prices scales by q, which proves Corollary 1.

For a general productivity shock, take fixed nonnegative reporting weights ϖ_{js} summing to one and write

$$c = \sum_{j,s} \varpi_{js} \log \widehat{a}_{js}, \qquad \widetilde{h}_{js} = \log \widehat{a}_{js} - c.$$

The full post-shock technology is $e^{\theta_s c} T_{js} \exp(\theta_s \widetilde{h}_{js})$, a common productivity scaling of the residual-shock technology by $q = e^c$. Under corresponding equilibrium selections, both shocks produce identical tariffs and identical log values of the system and its components. Real incomes and the level values differ by e^c. Reporting weights determine the decomposition but do not affect this equivalence.

B The policy game: existence, decomposition, and computation

This appendix gives sufficient conditions for the multi-country policy equilibrium, derives the finite decomposition used in the counterfactuals, and documents the computation.

B.1 Policy optimization and existence

This subsection states sufficient conditions for existence of a policy equilibrium. A pure-strategy equilibrium exists if tariff domains are nonempty, compact, and convex and each objective is quasi-concave in own tariffs, with welfare $W_i(\mathbf{T}, \mathbf{t})$ continuous under the maintained competitive-equilibrium selection. These conditions are not established for the calibrated model. The computations therefore assess candidate solutions using residual and unilateral-deviation checks.

For an independently adjustable tariff $t_{i\ell} \in [0, \bar{t}_{i\ell}]$, with $\bar{t}_{i\ell} > 0$, and writing $G_{i,t_{i\ell}} =$

$\partial G_i/\partial t_{i\ell}$ for the derivative of the objective in (1), necessary conditions are

$$\begin{cases} G_{i,t_{i\ell}} = 0, & 0 < t_{i\ell} < \bar{t}_{i\ell}, \\ G_{i,t_{i\ell}} \leq 0, & t_{i\ell} = 0, \\ G_{i,t_{i\ell}} \geq 0, & t_{i\ell} = \bar{t}_{i\ell}. \end{cases} \tag{B.1}$$

The derivatives include the competitive-equilibrium response. Concavity over the whole own domain is sufficient for these conditions to characterize a global best response. When a government sets one tariff for several partners, such as an MFN rate, bilateral rates are $\widetilde{\mathbf{t}}_i = \mathbf{B}_i \mathbf{t}_i + \mathbf{c}_i$ with a fixed matrix $\mathbf{B}_i$ and vector $\mathbf{c}_i$. The gradient with respect to the common tariff, $\mathbf{B}_i^{\mathsf{T}} \nabla_{\widetilde{\mathbf{t}}_i} G_i$, aggregates bilateral effects, and all gradients below refer to the tariffs a government actually sets.

Interior tariffs supply equality restrictions that identify welfare weights when the foreign-welfare margins have sufficient rank under the chosen parameterization. Tariffs at the bounds of positive-width domains, including zero tariffs, supply inequality restrictions. Singleton zero-ceiling domains supply no feasible direction. The estimator in Section 3.2 uses the interior equalities, while Section 3.3 evaluates the endpoint inequalities as diagnostics.

B.2 Decomposition of changes in value

This subsection defines the decomposition of Section 2.2 in the finite form that the tables report and records the two envelope facts behind its local form. The tools are standard, the chain rule and the envelope theorem for the local form and the sequential counterfactual for the finite form, and what is specific to the paper is the object decomposed, the value of an institution rather than welfare.

The local form holds along a branch on which the active constraints do not change and the first-order conditions of the interior tariffs have a nonsingular Jacobian, so that the implicit function theorem makes the equilibrium tariffs differentiable in z and the chain rule gives (4). Two envelope facts characterize its own-policy terms. At an interior noncooperative optimum the own-tariff gradient of own welfare is zero, and a tariff fixed at a bound has zero derivative, so the noncooperative own-policy term vanishes. Under cooperation, dividing the first-order condition of (1) by $P_i^0 W_i$ gives

$\partial u_i/\partial t_{i\ell} = -\sum_{j\neq i}\lambda_{ij}(P_j^0 W_j/P_i^0 W_i)\,\partial u_j/\partial t_{i\ell}$ for an interior tariff, which is generally nonzero because the government also values foreign welfare. One-sided derivatives may differ where a constraint switches between slack and binding, and the branch says nothing about discrete changes between branches.

The finite form replaces derivatives with differences between the six solved equilibria, the three regimes in the baseline and counterfactual productivity states, marked by superscripts 0 and 1, with weights, domains, and nontechnology primitives fixed within regimes and no regime-specific intercepts making the baseline tariffs optimal. Let $\mathbf{t}^{R,0}$ and $\mathbf{t}^{R,1}$ denote the equilibrium tariffs of regime R in the two states, with $\mathbf{t}_{-i}$ the tariffs of governments other than i, and $\bar{u}_i(\mathbf{T},\mathbf{t}) = \log W_i(\mathbf{T},\mathbf{t})$, which is u_i of Section 2.2 with the technology profile written explicitly, and define

$$\begin{aligned}
\Delta_i^{\text{exp},R} &= \bar{u}_i(\mathbf{T}^1,\mathbf{t}^{R,0}) - \bar{u}_i(\mathbf{T}^0,\mathbf{t}^{R,0}),\\
\Delta_i^{\text{par},R} &= \bar{u}_i(\mathbf{T}^1,\mathbf{t}_i^{R,0},\mathbf{t}_{-i}^{R,1}) - \bar{u}_i(\mathbf{T}^1,\mathbf{t}_i^{R,0},\mathbf{t}_{-i}^{R,0}),\\
\Delta_i^{\text{own},R} &= \bar{u}_i(\mathbf{T}^1,\mathbf{t}_i^{R,1},\mathbf{t}_{-i}^{R,1}) - \bar{u}_i(\mathbf{T}^1,\mathbf{t}_i^{R,0},\mathbf{t}_{-i}^{R,1}).
\end{aligned}$$

Here $\Delta_i^{\text{exp},R}$ is the exposure term at the initial tariffs, $\Delta_i^{\text{par},R}$ the effect of partners' tariff adjustment, and $\Delta_i^{\text{own},R}$ the effect of the government's own adjustment, each a competitive equilibrium at specified tariffs that need not be a policy equilibrium. They telescope to $\log W_i^{R,1} - \log W_i^{R,0}$, so

$$\Delta v_i = (\Delta_i^{\text{exp,sys}} - \Delta_i^{\text{exp,nc}}) + (\Delta_i^{\text{par,sys}} - \Delta_i^{\text{par,nc}}) + (\Delta_i^{\text{own,sys}} - \Delta_i^{\text{own,nc}}). \tag{B.2}$$

Reversing own and partner adjustments changes the attribution but not the total. The changes in v_i^{com} and v_i^{coop} are reported alongside in Table 4 and Appendix Table F1.

Two aggregates summarize the country values, the world log-gap index and the level measure,

$$100\,\frac{\sum_{i\in\mathcal{C}} I_i^0 v_i(\mathbf{T})}{\sum_{i\in\mathcal{C}} I_i^0} \qquad \text{and} \qquad V_{\mathcal{C}}(\mathbf{T}) = \sum_{i\in\mathcal{C}} P_i^0\Big[W_i^{\text{sys}}(\mathbf{T}) - W_i^{\text{nc}}(\mathbf{T})\Big],$$

where $\mathcal{C}$ contains the 34 named economies and the baseline-income weights are fixed across productivity states. The level measure, divided by baseline income, expresses the gain as a share of baseline modeled income and, unlike the index, moves with the level of

noncooperative income, so a common productivity increase scales it while leaving the index unchanged.

B.3 Computation and checks

Each regime is computed by projected Newton iteration on the stacked conditions (B.1). At each iteration all governments' gradients are evaluated at the same tariff profile. All tariffs then move simultaneously, using estimates of own curvature to scale the updates. A fixed point satisfies the first-order and bound conditions and is a candidate policy equilibrium. Damping stabilizes the iteration, alternative starts assess dependence on initialization, and the best-response searches below check for profitable unilateral deviations. These computations extend the tariff-equilibrium approaches of Ossa (2014) and Adão et al. (2026) to the specified ceiling constraints.

Each step recomputes the competitive equilibrium and the derivatives in (A.7), which are checked against finite differences of the re-solved equilibrium. Iteration stops when the largest projected relative objective gradient in log-gross-tariff coordinates is below 10^{-6}, with projection excluding nonpositive gradients at lower bounds and nonnegative gradients at upper bounds, and bound contact uses a tolerance of 10^{-7} in tariff fractions.

Each candidate undergoes projected quasi-Newton best-response searches for every government, each against the same candidate profile with the other governments held there, starting from the candidate and from the lower-bound schedule within the regime's domain. When some best responses improve on the candidate, their tariffs replace the candidate's together and the iteration restarts. These searches and residual checks provide numerical evidence of stationarity, without establishing global optimality or uniqueness.

Both productivity states are solved from the observed and lower-bound schedules. The climate state is also reached by continuation from the baseline solution. The level, finite-attribution, and ceiling-transition results retain this continuation benchmark. Climate tariff profiles differ across starts by as much as 13.55 percentage points, while the largest country-value difference between continuation and either direct start is 0.0054 log percentage points. The welfare results are therefore stable across these starts, although tariff-profile uniqueness is not established.

The baseline-state system equilibrium differs from observed tariffs. Table B1 reports the fit on interior cells and compares solved ceiling contacts with the observed ceiling

classification. The baseline value $v_i(\mathbf{T}^0)$ compares the solved system and noncooperative equilibria. The adjacent loss relative to observed tariffs provides a descriptive comparison.

Table B1: Numerical checks

	Baseline	Climate shock
Fit of the solved system to observed tariffs		
Root mean squared gap on interior cells, points	6.6	
agriculture	8.4	
extraction	4.5	
manufacturing	6.2	
Import share at binding ceilings, solved system	0.69	0.69
Import share in the observed ceiling class	0.84	
Convergence and optimality		
Largest first-order residual	9.94×10^{-7}	9.62×10^{-7}
Largest gain from a unilateral deviation	1.48×10^{-15}	
Smallest own-adjustment term, noncooperation		2.43×10^{-6}
Common productivity scale (Corollary 1)		
Tariffs, residual against full shock, points		$< 10^{-10}$
Values, residual against full shock		$< 10^{-11}$
Values, common component alone		$< 10^{-4}$

Notes: The root mean squared gap compares solved system tariffs with observed tariffs on interior cells. Ceiling shares use baseline imports on positive-width institutional domains as the denominator. Residuals are the largest projected relative first-order residual across the three regimes, against a stopping tolerance of 10^{-6}, and deviation gains are the largest relative gains found by best-response searches from the candidate and from the lower-bound schedule. The own-adjustment row is the smallest own-adjustment term across the 32 governments for the baseline-to-climate transition, partners first. The residual shock removes the common climate component (Appendix A.4). All 191 core and sensitivity profiles and 36 additional initialization and cap profiles meet both criteria, with a largest deviation gain of 1.93×10^{-7}.

Corollary 1 implies that the residual shock $\exp(\widetilde{h}_{js})$ reproduces the full shock's tariffs and log values under corresponding equilibrium selections. Table B1 reports the numerical differences. For the common component alone, the computation retains the baseline tariff profile and verifies its scaled counterpart using the same residual and unilateral-deviation criteria. This checks the corresponding selection required by the corollary.

The final check concerns enforcement. A government that discards both the welfare weight and the ceilings while its partners retain their system tariffs can gain. The value of the system therefore does not by itself establish that the system is self-enforcing. The gain from such a unilateral defection is

$$g_i = \max_{\mathbf{t}_i \in \mathcal{D}_i} \log W_i(\mathbf{T}, \mathbf{t}_i, \mathbf{t}_{-i}^{\text{sys}}) - \log W_i(\mathbf{T}, \mathbf{t}^{\text{sys}}). \tag{B.3}$$

The gain is nonnegative because the system tariff remains feasible. Median computed gains are 0.77 log percentage points for advanced economies and 0.39 for developing economies,

about 27 and 20 percent of each group's median value under the system. After the climate shock they are 0.77 and 0.40, so the shock leaves the incentive to defect essentially unchanged. The largest baseline gains accrue to Singapore, Argentina, Switzerland, and Israel. The comparison is static and excludes the repeated-game incentives needed to assess participation.

C Data and calibration

C.1 Sources and parameters

Bilateral trade flows are aggregated from the 2017 HS six-digit records of BACI in the HS2017 nomenclature, tariff schedules in other HS vintages are concorded to HS2017, and products are assigned to the three sectors through CPC 2.1 and ISIC Revision 4. The replication package records the concordances and their approximations.

Applied bilateral tariffs are from WITS, the preferential rate where one applies and the MFN rate otherwise, and the rest-of-world exporter pays MFN rates. Agricultural output uses FAOSTAT crop and livestock production values. Extraction and manufacturing use one-digit UNIDO INDSTAT output. Missing observations are supplemented with OECD TiVA 2025 gross output for 2017; Taiwan retains the available UNIDO observations. Section totals are not added to their constituent divisions. The base year is 2017.[8] The rest-of-world aggregate closes the model and is excluded from group summaries. Two design parameters complete the calibration: the 200 percent nominal cap on noninstitutional tariffs and the five-point margin below the ceiling that defines the interior sample. Table E2 reports alternatives to both.

The trade elasticities are the industry estimates of Caliendo and Parro (2015), except that three industries are assigned values chosen here, leather 5.00, basic and fabricated metals 5.81, and coke and refined petroleum 15.00 (Table C1). Equal weights across ISIC Revision 4 two-digit divisions give 8.11 for agriculture, 15.72 for extraction, and 5.62 for manufacturing.

[8]The 2017 schedule predates the additional United States tariffs introduced in 2018 and in April 2025.

Table C1: Industry elasticities and aggregation weights

Industry	ISIC divisions	Elasticity	Sector weight
Agriculture, forestry, and fishing	01–03	8.11	1
Mining and quarrying	05–09	15.72	1
Food, beverages, and tobacco	10–12	2.62	3/24
Textiles and apparel	13–14	5.60	2/24
Leather	15	5.00	1/24
Wood	16	10.83	1/24
Paper and printing	17–18	9.07	2/24
Coke and refined petroleum	19	15.00	1/24
Chemicals and pharmaceuticals	20–21	4.75	2/24
Rubber and plastics	22	1.67	1/24
Nonmetallic minerals	23	2.76	1/24
Basic and fabricated metals	24–25	5.81	2/24
Machinery	28	1.52	1/24
Computers, electronics, and electrical equipment	26–27	10.60	2/24
Transport equipment	29–30	1.84	2/24
Furniture, other manufacturing, and repair/installation	31–33	5.00	3/24

Notes: Elasticities are from Caliendo and Parro (2015) except the three reassignments named in the text. Each ISIC Revision 4 division receives its industry's elasticity, and the sector weight is the division's share of the divisions in the model sector, agriculture 01–03, extraction 05–09, and manufacturing 10–33.

C.2 The accounting bridge and the deficit purge

The bridge maps recorded trade and production into tariff-exclusive sales $M_{ji}^{s,0}$, production values B_{js}^0, and tariff-inclusive sector expenditures E_{is}^0. Domestic sales are production less exports, $M_{jj}^{s,0} = B_{js}^0 - \sum_{i \neq j} M_{ji}^{s,0}$. The raw accounts then satisfy

$$E_{is}^0 = \sum_j (1 + t_{ji}^{s,0}) M_{ji}^{s,0}, \qquad Y_i^0 = \sum_s B_{is}^0, \qquad I_i^0 = \sum_s E_{is}^0,$$

$$R_i^0 = \sum_{j \neq i, s} t_{ji}^{s,0} M_{ji}^{s,0}, \qquad D_i^0 = I_i^0 - Y_i^0 - R_i^0, \qquad \alpha_{is} = E_{is}^0 / I_i^0,$$

which express the baseline sourcing shares and producer-sales weights of Appendix A in bridge quantities as $\pi_{ji}^{s,0} = (1 + t_{ji}^{s,0}) M_{ji}^{s,0} / E_{is}^0$ and $\omega_{ji}^{s,0} = M_{ji}^{s,0} / Y_j^0$. All production is labor income in the model, so $Y_i^0 = w_i^0 L_i$ and the coverage of W_i^0 is the gross output of the three sectors. Raw gross output, value added, and employee compensation are distinct source concepts, and the bridge uses gross output throughout.

Two accounting reconciliations precede deficit removal. When recorded exports exceed 99.9 percent of available country-sector output, output is raised to leave domestic sales of at least one tenth of one percent of production. Equation (A.2) requires a positive

domestic share. European Union shipments within the union are domestic trade, consistent with its production total. Re-exports are retained as recorded. The replication package records the production sources and every floor adjustment.

Coverage remains incomplete. The agricultural source covers crops and livestock rather than the full agriculture, forestry, and fishing sector; it lacks Taiwan and the United Arab Emirates. The floor also raises agricultural output in Norway and Singapore and extraction output in Iran, Singapore, and the rest of the world. Rest-of-world production has missing constituent-country sources in all three sectors. The floor preserves the accounting system but does not identify the missing output or resolve source–export discrepancies. Results are conditional on this reconciliation.

Hong Kong is included in the rest-of-world aggregate. A separate accounting calculation that retains Hong Kong gives absorption of 4.33 times output after applying the domestic-sales floor. Among named economies, deficit removal raises Malaysia's real income by 22.5 log percentage points and lowers Pakistan's by 14.0, the largest increase and decrease.

The model imposes $D_i = 0$, so raw deficits are removed at observed tariffs and unchanged technologies by the hat-algebra purge of Dekle et al. (2008), with income $I_i = w_i L_i + R_i + D_i$ in the equations of Appendix A.2. Wages, expenditures, sourcing shares, and price indices adjust in the purge, and the purged equilibrium supplies the baseline objects $\pi_{ji}^{s,0}$, $\omega_{ji}^{s,0}$, and P_i^0 of Section 3.1.

The tariff supplement fills 12 missing United States copper-ore observations with a 2017 ad-valorem equivalent. It fills 36 missing crude-oil observations with a provisional 2019 ad-valorem equivalent of 2.4 percent. This cross-year proxy is a calibration assumption and covers about 37 percent of United States extraction imports.

C.3 Ceilings, preferential arrangements, and the interior sample

Bound rates are the final bound duties in the WTO Consolidated Tariff Schedules downloaded in August 2026, in each member's latest certified nomenclature and concorded to HS2017. They serve as the ceilings in force in 2017. The concessions date from the Uruguay Round or from each member's accession, later certifications are transpositions and rectifications, and lines still staging after 2017 are at most 4 percent of bound lines for every member but Kazakhstan. Its 2015 accession staged 27 percent of its lines to 2020, so its ceilings are somewhat tighter than its 2017 bindings, with little effect beyond

its own results.

Ceilings are available for 31 of the 33 WTO members in the sample. Saudi Arabia and the United Arab Emirates lack consolidated schedule files and belong to a customs union only partly represented. These two economies and the rest-of-world aggregate retain observed tariffs in every regime. Iran has no WTO ceiling and is excluded from estimation, but remains a counterfactual tariff setter under the pooled welfare weight.

Cell ceilings average product bound rates with the fixed weights in (C.1). Bound rates enter at the smaller of the rate and the cap. Explicitly unbound products enter at the cap. Products whose binding status the concordance does not resolve receive the cell's import-weighted average of known capped rates, or the cap if no product is known. Unresolved products carry 11.7 percent of active baseline imports and 54.7 percent of Switzerland's. Approximate concordance matches can be classified as known, so these shares do not capture every source of coverage error. A cell whose applied index exceeds its constructed bound index keeps the applied rate as its ceiling.

Preferential regimes come from the 2017 WITS preferential schedules, with regional groups expanded to members and the European Union pooled. A preferential ceiling equals the observed applied rate, so every preferential cell sits at its ceiling by construction and none enters the estimation sample. The baseline treats reciprocal agreements and unilateral schemes alike as commitments, although unilateral preferences can be revoked, and a sensitivity removes preferential ceilings while retaining the applicable WTO bindings.

Noninstitutional tariffs face a nominal cap of 200 percent, with one exception. Three Korean agricultural tariffs exceed it, 424.66 percent on imports from Argentina, 239.67 from Brazil, and 238.67 from South Africa, so their bounds in every regime equal these observed rates, which preserves feasibility without fixing the tariffs there. The three cells carry 0.0088 percent of active baseline imports. In the MFN sensitivity, Korea's nonpreferential agricultural partners share one additive shift that preserves the observed differences across their rates, and the cap constrains that shift rather than each rate. Korea's tariff on South African agricultural imports, for example, is 229.23 percent in the solved system of that sensitivity, its fixed offset above the solved shift, without the cap binding. Unilateral defection releases the institutional ceilings subject to these noninstitutional bounds.

An interior cell has a strictly positive applied rate at least δ percentage points below

its ceiling, with $\delta = 5$ in the baseline and two and ten points in sensitivities, a use of tariff water that relates to Beshkar et al. (2015) and Nicita et al. (2018). A corner is at zero or within δ points of its ceiling, and the classification is fixed across states and regimes. A cell's interior coverage ϱ^s_{ji} is the share of its baseline imports on products with tariffs strictly between zero and their ceilings. Table C2 reports the counts by economy and the average coverage over interior cells.

Table C2: Cell classification and interior coverage by economy

	Group	Interior	At zero	At ceiling	Interior coverage	Unknown
AUS	Adv.	8	41	51	0.14	0.05
CAN	Adv.	7	23	72	0.10	0.05
CHE	Adv.	0	44	56		0.55
CHL	Adv.	33	32	29	0.43	0.00
EU	Adv.	0	10	92		0.16
ISR	Adv.	51	16	19	0.34	0.01
JPN	Adv.	13	28	58	0.04	0.17
KOR	Adv.	16	14	72	0.64	0.03
NOR	Adv.	0	32	63		0.12
SGP	Adv.	0	80	0		0.04
TWN	Adv.	5	5	90	0.10	0.01
USA	Adv.	16	15	71	0.00	0.08
ARG	Dev.	33	12	27	0.55	0.15
BGD	Dev.	84	2	12	0.71	0.00
BRA	Dev.	43	9	45	0.66	0.08
CHN	Dev.	8	9	85	0.05	0.16
COL	Dev.	58	2	31	0.73	0.00
IDN	Dev.	65	7	29	0.52	0.13
IND	Dev.	43	1	58	0.62	0.03
IRN	Dev.	89	0	0	1.00	0.00
KAZ	Dev.	2	8	78	0.01	0.14
MEX	Dev.	33	24	21	0.49	0.05
MYS	Dev.	86	7	6	0.24	0.15
PAK	Dev.	74	0	19	0.99	0.00
PER	Dev.	37	35	17	0.20	0.00
PHL	Dev.	52	6	18	0.51	0.04
RUS	Dev.	2	7	92	0.12	0.16
THA	Dev.	89	9	4	0.33	0.10
TUR	Dev.	41	17	40	0.47	0.09
UKR	Dev.	2	8	63	0.00	0.01
VNM	Dev.	4	4	70	0.53	0.00
ZAF	Dev.	56	30	14	0.30	0.00
Total, 32 governments		1,050	537	1,402		

Notes: Cells are government-partner-sector cells with positive baseline imports. Advanced economies are those the World Bank classified as high income in 2017, with Taiwan treated as advanced, and the developing group is the rest. Interior coverage is averaged over the economy's interior cells, and Unknown is the import share of products whose binding status the concordance does not resolve. Iran's cells are interior because it has no schedule and are outside the estimation sample.

Of the 2,989 tariff cells, 961 are interior cells in the estimation sample, 1,402 are ceiling

cells, 135 are zero-tariff cells with positive-width domains, 402 are zero-tariff singleton domains (379 preferential and 23 nonpreferential), and 89 are Iran's interior cells, which lie outside the estimation sample because Iran's tariffs are not assumed to satisfy the estimating condition, although Iran remains a counterfactual player under the pooled weight. The broad regression uses the first three groups, zero tariffs and ceiling contacts supply the inequality diagnostics, and singleton domains supply no feasible direction and enter neither.

C.4 Estimation diagnostics

The baseline importer-clustered standard error is 0.063, with a 95 percent interval of 0.351 to 0.599. Allowing dependence between cells sharing a country in either role raises the standard error to 0.149 and gives an interval of 0.182 to 0.767. The instrumental-variable estimate is 0.373, with standard error 0.081 and clustered first-stage F statistic 81.4. The Anderson-Rubin 95 percent set is 0.185 to 0.510 (Anderson and Rubin, 1949). This clustered implementation uses asymptotic critical values; it does not establish exact coverage with finitely many importers.

Leave-one-importer estimates range from 0.443 without India to 0.506 without South Africa. Leave-one-exporter estimates range from 0.362 without the rest of world to 0.663 without China. Adding importer fixed effects gives 0.453, with standard error 0.068. Separate advanced- and developing-partner weights are 0.529 and 0.457. Their difference is 0.072, with importer-clustered standard error 0.066. Government-specific estimates for the sixteen governments with at least twenty interior cells, each from the government's own interior cells with exporter-clustered standard errors, range from -0.44 for Mexico to 1.08 for Peru, with twelve of the sixteen in $[0, 1]$. In the corresponding policy sensitivity, estimates are clipped to $[0, 1]$, and governments without sufficient identifying observations retain the pooled weight.

Without the preferential ceilings, the weight is 0.837 with an importer-clustered standard error of 0.087, so its 95 percent interval runs from 0.67 to 1.01. It lies more than three standard errors above the baseline estimate, which confirms that restraint the preferential ceilings absorbed is read as cooperation once they are removed, and it includes full internalization, consistent with the broad-sample estimate of 0.83. The instrumental-variable estimate in this specification is 0.764, inside that interval, so the

least-squares and instrument agreement of the baseline holds here as well.

Table 2 reports least-squares estimates under interior-coverage thresholds of one quarter and one half, and under interior-coverage weighting. The corresponding instrumental-variable estimates are 0.194 and 0.058, with standard errors 0.089 and 0.104, and interior-coverage weighting gives 0.185 with standard error 0.093. For the eight importers with at least twenty cells at every threshold, the unweighted mean of importer estimates falls from 0.407 at baseline to -0.002 at an interior coverage of one half. This is a descriptive comparison within common importers: the retained cells still change, so it does not distinguish aggregation error from sample composition. A censored likelihood that enters the zero-tariff cells of Table 2 as inequalities rather than equalities gives 0.56, the same as the equality treatment.

A further diagnostic allows violations of the estimating moments. Write $\langle\cdot\rangle$ for sample means over interior cells, Z for the free-trade instrument, and s_ε for the fitted residual root mean square, held fixed as a calibration scale. Imposing $|\langle X\varepsilon\rangle| \le c\sqrt{\langle X^2\rangle}s_\varepsilon$ yields the least-squares displacement bound $cs_\varepsilon/\sqrt{\langle X^2\rangle}$. The analogous IV bound is $c\sqrt{\langle Z^2\rangle}s_{\varepsilon,\mathrm{IV}}/|\langle ZX\rangle|$. At $c = 0.10$, these bounds are 0.051 and 0.056; at $c = 0.25$, they are 0.128 and 0.140. These are calibrated sensitivities to assumed moment violations, not estimates of bias, confidence bounds, or evidence that the identifying conditions hold.

C.5 Tariff aggregation and product incidence

For positive baseline imports, let ϕ^0_{jip} be the tariff-exclusive import share of product p within a bilateral-sector cell. The applied index and the ceiling index are

$$t^{s,0}_{ji} = \sum_{p\in s} \phi^0_{jip} t^0_{jip}, \qquad \bar{t}^s_{ji} = \sum_{p\in s} \phi^0_{jip} \bar{t}_{jip}, \qquad \phi^0_{jip} = \frac{M^0_{jip}}{\sum_{p'\in s} M^0_{jip'}}. \tag{C.1}$$

The applied index preserves baseline tariff revenue, although not price-index derivatives or revenue responses when import composition changes.

Interior cells contain constrained products. Across the 961 estimation cells, zero-tariff products carry 45.1 percent of imports and products with positive tariffs at or above their constructed ceilings carry 5.8 percent. Zero tariffs are classified first, so these categories are disjoint. Across all 1,050 interior cells, including Iran, the shares are 43.0 and 5.6 percent. These descriptive classifications are not legal compliance tests. Their shares

alone do not determine the direction of any bias in the estimated weight.

A product-level quadratic approximation provides a limited check on tariff aggregation. Each product's contribution is half the cell's effective elasticity $\theta_s(1 - \pi_{ji}^{s,0})$ times its share of baseline cell imports times the difference in squared log gross tariffs between the noncooperative tariff and the smaller of that tariff and the product ceiling. The cell version uses the cell ceiling instead. Across the 32 governments, the product sum is 1.70 percent of baseline income, compared with 1.68 at the cell level. This agreement concerns an aggregate approximation, not a product-level general-equilibrium validation. Within-cell dispersion in product ceilings is 11 points overall, 8 in advanced economies, and 15 in developing economies, so economy-level differences can be substantial.

D Productivity scenarios

This appendix documents the construction of the three counterfactual analyses of Section 5. Appendix D.1 builds the sector ranking and the shift behind the widening and narrowing of comparative advantage, Appendix D.2 calibrates the two concentrated shocks, and Appendix D.3 assembles the climate inputs by economy.

D.1 A capability ranking of sectors

The comparative-advantage experiment ranks sectors within each economy by cost competitiveness, using baseline sourcing shares and tariffs. An exporter's tariff-adjusted log spending share, relative to the partner's domestic share, depends on exporter capability, partner capability, and bilateral trade costs, where capability is technology net of the wage, $\kappa_{js} = \ln T_{js} - \theta_s \ln w_j$. Comparing both directions of trade removes bilateral trade costs when they are symmetric within the sector, as in the ratio of Head and Ries (2001). Each pair with positive trade in both directions then gives an equation in capability differences, and least squares solves these equations with capabilities normalized to sum to zero within each sector. The rest of the world is excluded from the ranking and from the corresponding productivity shifts. The index

$$C_{js} = \exp\left\{\frac{\widehat{\kappa}_{js} - \sum_m \nu_m^s \widehat{\kappa}_{ms}}{\theta_s}\right\}, \tag{D.1}$$

with fixed baseline export weights ν_m^s summing to one within each sector, converts the estimated capability into productivity units relative to the sector's export-weighted mean. It measures a_{js}/w_j relative to that mean and is invariant to how each sector is normalized, so it ranks cost competitiveness across sectors within an economy.

Widening raises productivity by a common log shift of 0.046 in each economy's highest-ranked sector and lowers it by the same amount in its lowest-ranked sector, and narrowing applies the opposite changes, which can reverse a ranking when the initial gap is small. The shift is of the order of two decades of relative convergence. Levchenko and Zhang (2016) estimate that about 18 percent of the within-country gap in relative sectoral productivity closes per decade, and the log gap between the most and least competitive sectors in Table D1 has a median of 0.25 across the 32 governments, so one decade at that rate moves each extreme sector of the median economy by about 0.022 log points. The common shift therefore gives an approximate historical scale rather than imposing the same convergence rate on each economy. Revealed export specialization provides the alternative ranking used in Section 6.

Table D1: Competitiveness index by economy

	Agr.	Extr.	Manuf.		Agr.	Extr.	Manuf.
AUS	0.95	**1.00**	*0.77*	IDN	**0.99**	0.95	*0.87*
CAN	0.97	**1.01**	*0.81*	IND	**1.02**	*0.90*	0.95
CHE	*0.69*	**0.80**	0.75	IRN	**0.79**	0.78	*0.68*
CHL	0.98	**1.00**	*0.67*	KAZ	0.76	**1.02**	*0.61*
EU	1.01	*0.95*	**1.13**	MEX	0.93	**1.04**	*0.75*
ISR	0.81	**0.89**	*0.72*	MYS	0.80	**0.84**	*0.78*
JPN	0.77	*0.74*	**1.06**	PAK	0.91	**0.92**	*0.78*
KOR	*0.65*	0.70	**0.96**	PER	0.96	**1.00**	*0.66*
NOR	*0.53*	**1.00**	0.69	PHL	**0.85**	0.69	*0.68*
SGP	*0.41*	0.62	**0.76**	RUS	0.93	**1.07**	*0.86*
TWN	*0.39*	0.62	**0.87**	THA	**0.90**	*0.84*	0.85
USA	**1.11**	*1.07*	1.08	TUR	0.85	**0.93**	*0.81*
ARG	**1.15**	0.97	*0.85*	UKR	**1.01**	0.92	*0.71*
BGD	0.71	*0.71*	**0.74**	VNM	0.86	**0.89**	*0.85*
BRA	**1.17**	0.92	*0.90*	ZAF	0.94	**1.05**	*0.72*
CHN	1.04	*0.91*	**1.24**	SAU	*0.66*	**1.10**	0.74
COL	1.00	**1.03**	*0.64*	ARE	*0.40*	**1.02**	0.64

Notes: The index is (D.1) with baseline export weights. In each row, bold marks the sector that widening raises and italics the sector it lowers, and narrowing does the reverse. The alternative ranking by revealed export specialization, used in Table F1, places a different sector highest for twelve economies, Canada, Chile, India, Indonesia, Iran, Israel, Malaysia, Mexico, Pakistan, Switzerland, Turkey, and Vietnam.

The rank-preserving proportional sensitivity of Section 6 sets $\log \widehat{a}_{js} = \rho(c_{js} - \bar{c}_j)$ for widening, where $c_{js} = \log C_{js}$, $\bar{c}_j$ is its mean across the economy's sectors, and

$\rho = 2(0.046)/\operatorname{median}_j(\max_s c_{js} - \min_s c_{js}) \simeq 0.373$ matches the fixed experiment's change in the median extreme-sector gap across ranked governments. Narrowing reverses the sign. The shift preserves each economy's geometric-mean productivity and scales every within-economy log capability gap by $1 + \rho$ or $1 - \rho$ without reversing rankings, and the rest of world is unshocked.

D.2 Calibration of the concentrated shocks

The Chinese contraction reverses the historical expansion of Chinese manufacturing productivity. Hsieh and Ossa (2016) estimate median annual growth in fundamental productivity across Chinese manufacturing industries of 3.5 to 3.7 percent over 1995 to 2007, and reversing twelve years at these rates lowers productivity by 33.8 to 35.3 percent, which brackets the 35 percent used in Section 5.2. The 20 percent sensitivity matches the post-2007 slowdown, since the growth rates of 4.5 percent over 1998 to 2007 and 1.1 percent over 2007 to 2013 in Brandt et al. (2026) leave productivity 18 percent below its continued-growth path after six years. The figures are industry medians, so a single multiplier is a historically scaled experiment rather than a reconstruction, and it applies to productive efficiency, $\widehat{a}_{\text{CHN,man}} = 0.65$ with $\widehat{T} = \widehat{a}^{\,\theta}$.

The United States extraction expansion matches measured productivity growth in mining during the shale era. The Bureau of Labor Statistics' industry-level total factor productivity accounts report mining productivity (NAICS 21, including oil and gas extraction, other mining, and support activities) 26.5 percent higher in 2017 than in 2007, and oil and gas extraction alone (NAICS 211) 64.5 percent higher. The measured change is growth during the shale era rather than shale's causal contribution, and the 2017 baseline already contains it, so the shock is an additional expansion of historical magnitude, with the 50 and 80 percent cases as sensitivities. Modeling shale as extraction efficiency follows Arezki et al. (2017), who find that an efficiency-units representation and an endowment representation have virtually the same implications.

D.3 Climate inputs and source mappings

The climate scenario takes its productivity inputs from Nath (2025), who assembles country-level projections of climate damage to agricultural and manufacturing productivity under RCP 8.5 for 2080 to 2099. For agriculture, each economy's multiplier is the arithmetic

mean of its available multipliers from three of his four projections, the Climate Impact Lab estimate (Hultgren et al., 2025), Cline (2007), which includes carbon fertilization, and the crop-model projections he takes from Iglesias and Rosenzweig, excluding the crop-model input of Costinot et al. (2016), and the source-specific scenarios of Table E2 use one projection at a time. For manufacturing, the multiplier is one plus his projected proportional change in revenue per worker. Both enter the technology parameter as $\widehat{T}_{is} = (\widehat{a}_{is})^{\theta_s}$, extraction is unshocked, and the projections are used as assembled, without re-estimation. Table D2 reports the resulting vector, with the proxy and regional assignments in its notes. Its common component is a decline of 2.36 log percentage points under expenditure weights, which Corollary 1 removes from proportional value, so only the relative changes, concentrated in agriculture, can move it.

Table D2: The climate shock by economy

	Agriculture	Manufacturing		Agriculture	Manufacturing
AUS	−24.2	0.0	IND	−17.5	0.0
CAN	−6.2	−0.7	IRN	−19.2	0.2
CHE	−4.5	−3.1	KAZ	−0.7	−3.1
CHL	−10.9	−0.9	MEX	−16.5	0.0
EU	−9.0	−1.3	MYS	−9.5	0.0
ISR	−24.0	0.0	PAK	−22.1	0.1
JPN	0.4	−1.0	PER	−16.2	−0.5
KOR	19.9	−1.5	PHL	−9.6	0.0
NOR	11.1	−0.6	RUS	−17.0	−0.6
SGP	−13.5	0.0	THA	−17.5	0.0
TWN	−6.2	−3.6	TUR	−14.9	−3.8
USA	−12.4	0.3	UKR	−13.7	−3.8
ARG	−9.8	0.0	VNM	−6.3	0.0
BGD	−4.7	−1.6	ZAF	−23.1	0.0
BRA	−4.5	0.0	SAU	−19.2	0.2
CHN	−6.2	−3.6	ARE	−19.2	0.2
COL	−13.0	0.0	RoW	−19.1	−2.7
IDN	−9.8	0.0			

Notes: Productivity changes $100(\widehat{a}_{is} - 1)$ of the main climate scenario, from the replication materials of Nath (2025), with extraction unshocked. Taiwan uses China's input, Saudi Arabia and the United Arab Emirates use Iran's, and the European Union and the rest of the world use pooled aggregates.

E Additional tables

This appendix collects the additional tables cited in the main text and in Appendices C and D. Table E1 crosses the preferential ceilings with the estimated weight. Table E2 reports the estimation, policy, calibration, and climate-input variants. Table E3 gives

country values at baseline and climate productivity and their exposure and policy attribution. Table E4 reports mean tariffs by regime, group, and sector. Table E5 reports each restraint's stand-alone value and the incidence of the concentrated shocks. Table E6 reports the value of each restraint alone under alternative weights, and Table E7 reports alternative shock constructions and aggregation checks.

Table E1: Preferential ceilings and the estimated weight

		Preferential ceilings retained		Preferential ceilings removed	
Weight specification	λ	Baseline	Climate	Baseline	Climate
Baseline estimate	0.475	0.810	0.804	0.769	0.763
Re-estimated without preferential ceilings	0.837	0.838	0.835	0.836	0.832

Notes: Full-system values in log percentage points relative to noncooperation at baseline and at climate productivity. Removing the preferential ceilings retains the WTO bound rates. The weight of 0.837 (standard error 0.09) is estimated on the interior sample after the preferential ceilings are removed.

Table E4 reports tariffs by regime. Noncooperative means are close to 16 percent in each sector, and no noncooperative tariff reaches the numerical cap. Their limited sectoral variation is consistent with the uniform-tariff intuition in one-factor Ricardian models (Opp, 2010; Costinot et al., 2015). The levels are below the Nash tariffs of about 60 percent in Ossa (2014), whose model has lower trade elasticities and markup distortions. The largest sectoral fit discrepancies are in agriculture, where system means are below observed means in both groups.

Table E2: The value of the system under alternative specifications

	Weight	World value	Climate response	
Specification	λ	Baseline	World	Median
Baseline	0.475	0.810	−0.0061	−0.025
Estimation and welfare weights				
Lower confidence endpoint	0.351	0.792	−0.0061	−0.023
Upper confidence endpoint	0.599	0.825	−0.0059	−0.029
Interior coverage at least one half	0.209	0.772	−0.0061	−0.025
Lower confidence endpoint of the one-half coverage weight	0.039	0.747	−0.0058	−0.028
Government-specific weights	$\{\widehat{\lambda}_i\}$	0.803	−0.0059	−0.023
Margins not scaled	0.674	0.832	−0.0057	−0.028
Margins scaled by cell imports	0.509	0.815	−0.0061	−0.025
Iran included in the estimation sample	0.404	0.801	−0.0061	−0.026
Policy domains and calibration				
MFN sensitivity	0.523	0.812	−0.0101	−0.035
Preferential ceilings removed	0.837	0.836	−0.0039	−0.056
Interior classification margin: 2 points	0.476	0.810	−0.0061	−0.025
Interior classification margin: 10 points	0.464	0.809	−0.0061	−0.025
Domain and construction cap: 150 percent	0.475	0.811	−0.0061	−0.025
Domain and construction cap: 300 percent	0.475	0.810	−0.0061	−0.025
Trade elasticities multiplied by 0.8	0.563	0.995	−0.0047	−0.027
Trade elasticities multiplied by 1.2	0.389	0.684	−0.0077	−0.032
Saudi Arabia and UAE as tariff setters	0.479	0.821	−0.0055	−0.027
Climate components and sources, weight fixed				
Agricultural component alone	0.475	0.810	−0.0018	−0.007
Manufacturing component alone	0.475	0.810	−0.0049	−0.021
Climate Impact Lab agricultural input	0.475	0.810	−0.0044	−0.001
Cline agricultural input	0.475	0.810	0.0094	0.007
Crop-model agricultural input	0.475	0.810	−0.0042	−0.032
Taiwan assigned South Korea's climate inputs	0.475	0.810	−0.0059	−0.025

Notes: Log percentage points, with medians over tariff-setting governments. Except where the weight is stated as fixed, each specification re-estimates the pooled weight. Confidence endpoints use importer-clustered 95 percent intervals, government-specific weights are clipped to $[0, 1]$, and removing preferential ceilings retains the WTO bound rates. The margins are the marginal welfare effects of a cell's tariff in levels at baseline prices, which the baseline divides by the importer's baseline income, the row with margins not scaled divides by nothing, and the row scaled by cell imports divides by the cell's baseline imports. The cap rows change the numerical domain and ceiling construction together, and changing either alone moves world value by less than 0.001 log percentage points. The three agricultural-source rows replace the mean of three projections with a single projection, retaining the main-scenario multiplier where that source has no observation (Appendix D.3).

Table E3: The value of the system by economy under the climate shock

	Group	Fixed	System	Noncoop.	Value, shock	Value, baseline	Exposure	Policy
AUS	Adv.	−1.97	−2.09	−1.83	2.82	3.08	−0.24	−0.02
CAN	Adv.	−0.69	−0.55	−0.60	4.14	4.10	0.05	0.00
CHE	Adv.	−2.79	−2.81	−2.88	5.10	5.03	0.05	0.02
CHL	Adv.	−1.74	−1.72	−1.63	3.01	3.10	−0.10	0.00
EU	Adv.	−1.62	−1.61	−1.62	0.37	0.36	0.00	0.01
ISR	Adv.	−1.57	−1.56	−1.61	2.65	2.60	0.02	0.03
JPN	Adv.	−0.94	−1.00	−1.00	0.47	0.47	−0.01	0.01
KOR	Adv.	−0.98	−1.66	−1.65	1.90	1.92	0.00	−0.01
NOR	Adv.	0.12	0.10	0.26	4.99	5.14	−0.02	−0.13
SGP	Adv.	−0.21	−0.21	−0.23	3.78	3.76	−0.01	0.03
TWN	Adv.	−3.48	−3.47	−3.51	1.68	1.63	0.03	0.02
USA	Adv.	−0.48	−0.46	−0.43	0.51	0.54	−0.04	0.01
ARG	Dev.	−1.29	−1.37	−1.41	0.38	0.34	0.10	−0.07
BGD	Dev.	−2.27	−2.39	−2.35	2.09	2.14	0.00	−0.05
BRA	Dev.	−0.68	−0.66	−0.59	0.99	1.05	−0.08	0.02
CHN	Dev.	−3.66	−3.66	−3.67	0.15	0.14	0.02	0.00
COL	Dev.	−2.85	−2.85	−2.74	1.88	1.98	−0.11	0.00
IDN	Dev.	−1.88	−1.85	−1.79	0.76	0.82	−0.06	0.00
IND	Dev.	−4.35	−4.27	−4.26	0.32	0.34	−0.01	−0.01
IRN	Dev.	−3.76	−3.80	−3.72	2.38	2.46	−0.06	−0.02
KAZ	Dev.	−1.60	−1.57	−1.60	3.25	3.22	0.07	−0.04
MEX	Dev.	−1.68	−1.66	−1.63	6.24	6.27	−0.05	0.03
MYS	Dev.	−1.14	−1.13	−1.08	4.21	4.26	−0.08	0.02
PAK	Dev.	−7.19	−7.22	−7.11	0.84	0.96	−0.03	−0.08
PER	Dev.	−2.34	−2.34	−2.24	1.52	1.62	−0.10	0.00
PHL	Dev.	−2.31	−2.23	−2.15	4.02	4.11	−0.08	0.00
RUS	Dev.	−2.13	−2.20	−2.14	1.40	1.45	−0.04	−0.01
THA	Dev.	−1.38	−1.31	−1.34	2.46	2.43	−0.01	0.04
TUR	Dev.	−5.16	−5.35	−5.41	1.05	0.99	0.10	−0.03
UKR	Dev.	−5.44	−5.17	−5.27	2.01	1.91	0.12	−0.02
VNM	Dev.	−0.73	−0.67	−0.65	1.84	1.87	−0.02	0.00
ZAF	Dev.	−2.44	−3.10	−2.90	1.74	1.94	−0.23	0.02
SAU	Nonplayer	−1.23	−1.25	−1.15	2.25	2.36	−0.10	−0.01
ARE	Nonplayer	−0.60	−0.53	−0.49	3.12	3.16	−0.09	0.04

Notes: Log percentage points of real income under the climate shock of Section 5.3. The first three columns are the shock's welfare change at the fixed observed schedule, under the system, and under noncooperation, with regime tariffs solved in both states. Value is $100v_i$ at the shocked and baseline technologies, and exposure and policy are its partners-first finite attribution, with policy summing both policy terms.

Table E4: Mean tariffs by regime, group, and sector

Importer group	Sector	Observed	System	Commit.	Noncoop.
Advanced	agriculture	8.2	1.9	3.0	16.2
Advanced	extraction	0.4	0.6	1.0	16.3
Advanced	manufacturing	2.3	1.8	2.6	16.2
Developing	agriculture	7.1	4.3	8.0	15.7
Developing	extraction	2.4	3.7	6.7	15.8
Developing	manufacturing	6.3	4.3	7.8	15.7

Notes: Unweighted mean applied tariffs, in percent, across cells with positive baseline trade in each importer group and sector. Every mean changes by less than a quarter of a percentage point under the climate shock.

Table E5: Value of each restraint alone and changes under the concentrated shocks, by economy

	Baseline value			Change in value	
	Commitments only	Cooperation only	Full system	China contracts	United States expands
Advanced economies					
AUS	2.88	1.41	3.08	−1.73	−0.64
CAN	4.01	1.70	4.10	−0.19	−1.31
CHE	5.10	2.29	5.03	0.35	−0.10
CHL	3.15	1.41	3.10	−0.86	−0.54
EU	0.22	0.72	0.36	−0.01	0.03
ISR	2.55	1.74	2.60	−0.11	0.01
JPN	0.35	0.70	0.47	0.18	0.06
KOR	1.86	1.71	1.92	0.25	0.08
NOR	5.02	1.37	5.14	−0.84	−1.23
SGP	3.49	2.32	3.76	0.12	−0.14
TWN	1.52	1.40	1.63	0.28	−0.02
USA	0.46	0.90	0.54	−0.13	0.69
Developing economies					
ARG	0.21	1.00	0.34	−0.17	−0.09
BGD	1.83	1.65	2.14	−0.04	0.09
BRA	1.03	0.58	1.05	−0.31	−0.12
CHN	0.14	0.35	0.14	0.36	−0.02
COL	1.98	0.87	1.98	−0.40	−0.55
IDN	0.71	0.66	0.82	−0.36	−0.11
IND	0.33	0.53	0.34	0.02	0.05
IRN	2.45	1.04	2.46	−1.07	−0.41
KAZ	3.13	1.32	3.22	−0.62	−0.64
MEX	6.26	2.17	6.27	−0.15	−0.80
MYS	4.34	2.04	4.26	−0.19	−0.12
PAK	0.71	0.74	0.96	−0.19	−0.01
PER	1.61	0.81	1.62	−0.75	−0.39
PHL	3.95	2.14	4.11	−0.32	−0.20
RUS	1.34	0.78	1.45	−0.46	−0.35
THA	2.45	1.55	2.43	0.09	−0.01
TUR	1.00	1.22	0.99	0.07	0.07
UKR	1.58	1.13	1.91	−0.26	−0.02
VNM	1.67	1.41	1.87	−0.16	−0.01
ZAF	1.67	1.18	1.94	−0.49	−0.14

Notes: Log percentage points of real income for the 32 optimizing governments. The first three columns are values relative to noncooperation at 2017 productivity under each restraint alone and under both. The last two are the changes in the value of the full system under the Chinese manufacturing contraction and the United States extraction expansion, in points of each economy's real income.

Table E6: The value of each restraint alone under alternative weights

			Share of the value (%)	
Weight specification	λ	Value	Cooperation alone	Commitments alone
Illustrative lower weight	0.260	0.779	68.3	95.2
Lower confidence endpoint	0.351	0.792	81.8	93.6
Free-trade instrument	0.373	0.796	84.5	93.2
Baseline estimate	0.475	0.810	94.0	91.6
Upper confidence endpoint	0.599	0.825	100.9	89.9

Notes: Baseline productivity and baseline ceilings. Value is the full-system value in log percentage points relative to noncooperation, and the last two columns report the value under each restraint alone as a percentage of it at the same weight. The confidence endpoints are the importer-clustered 95 percent interval, and the illustrative weight of 0.26 lies below it.

Table E7: Sensitivity to shock construction and aggregation

	World	Median	Lower weight	Upper weight
A. Changes in value under alternative shock constructions and weights				
Fixed shift, widening	2.23	11.47	2.29	2.35
Fixed shift, narrowing	−0.10	−7.37	−0.14	−0.17
Proportional shift, widening	3.18	7.48	3.16	3.36
Proportional shift, narrowing	−1.32	−6.80	−1.25	−1.43
Chinese manufacturing contraction	6.68	−8.10	6.92	6.34
United States extraction expansion	3.13	−4.27	3.14	3.12
Climate change	−0.76	−1.29	−0.77	−0.71
	Equal weights			
B. Changes in aggregate value under equal country weights				
Chinese manufacturing contraction	−12.61			
United States extraction expansion	−10.22			
Climate change	−1.59			

Notes: Percent changes in the value of the system under the baseline ceilings, with aggregate changes divided by the aggregate baseline value under the same weights and medians over the 32 governments' changes relative to their own baselines. In Panel A the first two columns use the baseline weight and the last two the importer-clustered 95 percent endpoints, 0.351 and 0.599, as sensitivity inputs. Proportional shifts preserve sector rankings (Appendix D.1). Panel B changes only the reporting weights, to equal weights across the 34 named economies.

F Additional productivity shifts

This appendix reports the alternative sector rankings, shock magnitudes, and sign reversals discussed in Sections 5 and 6, collected in Table F1.

Ranking sectors by revealed export specialization preserves the signs of the aggregate changes but changes their magnitudes. Widening raises world value by 0.029 rather than 0.018 log percentage points, and the reverse shift lowers it by 0.009 rather than 0.001. The restraint pattern of Table 4 holds for widening under both rankings, with commitments alone changing almost as much as the total and cooperation alone by less, but not for narrowing, where cooperation alone moves more than commitments alone under both rankings. The median widening effect also changes sign, from 0.25 to -0.01 log percentage points. The aggregate direction is therefore more stable than its attribution or country incidence.

Aggregate value increases more than proportionally across the reported concentrated-shock magnitudes. The 20 percent Chinese contraction raises world value by 0.009 log percentage points, compared with 0.054 at 35 percent. The United States extraction expansions raise it by 0.025, 0.103, and 0.287 points at 27, 50, and 80 percent. Binding ceilings rise from 909 to 944 across those three cases. At 80 percent, tariff responses reinforce the exposure term, and at the two smaller magnitudes they offset it. Under the 20 percent Chinese contraction the two restraints move in opposite directions, commitments alone up by 0.015 log percentage points and cooperation alone down by 0.009, so the direction of cooperation's response need not follow the total. The main 27 percent experiment follows the historical calibration in Appendix D.2. Reversing the shocks changes their incidence. Aggregate value rises under the Chinese expansion while China's own value changes little. It falls under the United States extraction contraction, which reduces the United States' own value by about 46 percent.

Table F1: Additional productivity shifts

	Change,	Change in the value		Commitments	Cooperation	Ceilings
	tariffs fixed	World	Median	alone	alone	binding
Alternative sector ranking						
Widening, capability ranking	2.41	0.018	0.25	0.019	0.006	875
Widening, revealed ranking	2.77	0.029	−0.01	0.026	0.020	879
Narrowing, capability ranking	−2.13	−0.001	−0.14	0.001	−0.005	958
Narrowing, revealed ranking	−2.52	−0.009	0.00	−0.005	−0.014	961
Alternative magnitude						
China, manufacturing down 35 percent	−12.91	0.054	−0.18	0.054	0.015	926
China, manufacturing down 20 percent	−6.81	0.009	−0.10	0.015	−0.009	926
United States, extraction up 27 percent	0.41	0.025	−0.10	0.023	0.010	909
United States, extraction up 50 percent	1.01	0.103	0.00	0.101	0.054	923
United States, extraction up 80 percent	1.95	0.287	0.20	0.284	0.151	944
Sign reversal						
China, manufacturing up 35 percent	9.72	0.024	0.05	0.014	0.018	887
United States, extraction down 27 percent	−0.15	−0.007	0.03	−0.007	−0.007	907

Notes: The first row of each panel is a scenario of Table 4, reported there in percent of baseline value and here in log percentage points, with world and median the world change and the median country change. Tariffs-fixed welfare changes use the observed schedule. Commitments alone and cooperation alone are the world changes in the value of each restraint alone (Appendix B.2), and the last column counts covered tariffs at their system ceilings after the shift, against 916 initially.